Edge-Aware Multimodal Emotion Recognition in IoT Environments

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James Thomas Black

B00288366

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Supervisor:

Professor Muhammad Zeeshan Shakir

School of Computing, Engineering and Physical Sciences

University of the West of Scotland

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3. **Journal Article (Under Review)**

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   Publisher: Springer Nature

4. **Conference Paper (Published)**

   Title: IoT-enabled Edge Architecture for Real-Time Facial Emotion Recognition

Authors: James Thomas Black and Muhammad Zeeshan Shakir

Conference: IEEE Symposium Series on Computational Intelligence

Publisher: IEEE

Author's Declaration

I declare that this thesis was written by me, and unless otherwise expressly stated in the article, the works contained in this article is my own and have not submitted for any other degree or professional qualifications. This is an exact copy of the thesis, including any final revisions required by my examiners. I understand that my thesis may be made available to the public electronically. I understand the ethical implications of my research.We have properly cited and referenced the original sources whenever we have used materials (data, theoretical analysis, and text) from other sources in the text of the report. I also declare that I have adhered to all principles of academic honesty and integrity and that I have not misrepresented, fabricated, or falsified any idea/data/fact/source in my submission and that I have followed the Institute norms and guidelines, as well as the regulations outlined in the Institute's Ethical Code of Conduct.

Name: **James Thomas Black**

Banner ID: **B00288366**

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Abstract

This thesis presents a scalable, low-cost, and privacy-conscious framework for real-time emotion recognition using Internet of Things (IoT) technologies and machine learning. Traditional systems often rely on high-cost, invasive hardware and are typically limited to controlled environments. In contrast, this research demonstrates that high-performance emotion classification can be achieved using lightweight models and affordable sensors suitable for deployment on edge devices.

Three primary objectives were addressed: the evaluation of unimodal classifiers across text, speech, and thermal image data; the development of a reproducible data collection framework using low-cost thermal imaging hardware; and the implementation of a real-time multimodal emotion recognition (MER) system optimised for embedded platforms. A novel thermal facial expression dataset was introduced, collected using an IoT-based thermal camera across six emotions and five national backgrounds. For textual emotion recognition, traditional machine learning models combined with TF-IDF features achieved up to 90% accuracy, outperforming transformer-based models such as DistilBERT in both efficiency and latency when deployed on edge hardware. In the thermal domain, a modified Vision Transformer (ViT) achieved 96% classification accuracy, surpassing ResNet in both accuracy and generalisation across hybrid datasets. The final Multimodal Emotion Recognition system, combining thermal, speech, and text modalities using both feature-level and decision-level fusion, demonstrated robust real-time performance. Decision-level fusion achieved up to 99% accuracy on structured datasets, with inference latency below 0.8 seconds and throughput exceeding 4,200 samples per second on a Raspberry Pi 4B.

These results confirm that accurate, real-time emotion recognition is achievable on affordable hardware without compromising performance. The system developed in this research offers a practical, deployable framework for emotion-aware computing in real-world applications such as healthcare, education, and human-computer interaction.

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Acronyms

BERT Bidirectional Encoder Representations from Transformers

BoW Bag of Words

CNB Complement Naive Bayes

CNN Convolutional Neural Network

DistilBERT Distillated-Bidirectional Encoder Representations from Transformers

DL Deep Learning

DT Decision Tree

FER Facial Expression Recognition

GNB Gaussian Naive Bayes

IEMOCAP Interactive Emotional Dyadic Motion Capture

IoT Internet of Things

KNN K-Nearest Neighbors

KTFE Kotani Thermal Facial Expression

LR Logistic Regression

LSTM Long Short-Term Memory

MELD Multimodal EmotionLines Dataset

MER Multimodal Emotion Recognition

ML Machine Learning

MNB Multinomial Naive Bayes

NLP Natural Language Processing

NVIE Natural Visible and Infrared Facial Expression Database

RF Random Forest

RGB Red Green Blue

SGD Stochastic Gradient Descent

SVM Support Vector Machines

TF-IDF Term Frequency-Inverse Document Frequency

UWSTF University of the West of Scotland Thermal Faces

ViT Vision Transformer

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Chapter 1

Introduction

In recent years, emotion recognition has become an increasingly important area of research, with applications across a wide range of domains including healthcare, education, automotive systems, and human-computer interaction. These systems rely on the accurate identification of human emotional states, often through the analysis of facial expressions, voice, physiological signals, or behavioural patterns. Traditionally, such capabilities have depended on expensive and often invasive hardware, limiting their deployment to controlled or commercial settings. However, the rapid expansion of the Internet of Things (IoT) and the increasing availability of low-cost sensors have opened new opportunities for creating scalable, accessible, and ethically responsible emotion recognition systems.

The potential of low-resolution and non-invasive sensors such as thermal cameras, when paired with recent advances in Deep Learning (DL) architectures, presents a compelling research challenge. Unlike high-resolution visible imagery, thermal data does not compromise user privacy and can still provide valuable physiological cues associated with emotional states. Meanwhile, the development of Convolutional Neural Networks (CNNs) and more recently, Vision Transformers (ViTs), has transformed the field of image-based classification by improving performance on tasks involving limited or noisy data. Yet, the application of these models in real-world, resource-constrained environments - especially in the context of thermal or Multimodal Emotion Recognition Multimodal Emotion Recognition (MER) - remains underexplored.

This thesis investigates the feasibility of using affordable, IoT-based thermal imaging devices for emotion classification and explores the effectiveness of DL models in both unimodal and multimodal contexts. By introducing a new thermal facial expression dataset, comparing the performance of CNN and ViT-based architectures, and evaluating model generalisation through hybrid datasets, this research seeks to advance the practical deployment of emotion recognition systems. Furthermore, it aims to contribute towards the broader field of affective computing by addressing limitations in current methodologies, particularly with regard to cost, accessibility, and data diversity.

The remainder of this chapter is structured as follows: Section 1.1 outlines the background and motivation for the research. Section 1.2 presents the key research objectives. Section 1.3 describes the progression of the PhD journey, detailing the development of each research output. Finally, Section 1.4 summarises the structure and organisation of the thesis.

1.1 Background and Motivation

Emotion recognition has become a critical component in enabling intelligent systems to interpret and respond to human behaviour. It plays a central role in applications such as adaptive learning, virtual assistants, driver monitoring systems, and mental health support tools. These systems typically rely on non-verbal cues, particularly facial expressions, to classify emotional states. Historically, the field has been dominated by approaches that depend on high-resolution visible spectrum cameras and expensive, specialist hardware. While effective in controlled environments, such solutions raise concerns around cost, scalability, and privacy, making them unsuitable for many real-world applications.

With the proliferation of IoT technologies, the landscape of sensing and data collection has begun to shift. Low-cost, portable, and power-efficient sensors are now readily available, allowing emotion recognition systems to be embedded into everyday environments. This shift introduces new opportunities for deploying affective computing in practical, cost-sensitive, and privacy-aware settings. In particular, thermal imaging has emerged as a compelling alternative to visible light-based systems. Thermal data, by capturing physiological changes such as increased blood flow or temperature variations associated with emotional arousal, can offer meaningful insights while simultaneously mitigating privacy risks, as it does not capture personally identifiable facial features.

Despite these advantages, emotion recognition using thermal imagery - especially in combination with other modalities - remains underexplored. Most existing research has focused on unimodal data or relied on costly thermal sensors, limiting the development of accessible, generalisable solutions. Moreover, many models currently in use are computationally intensive and not optimised for deployment on resource-constrained IoT devices.

DL has driven significant improvements in classification accuracy for Facial Expression Recognition (FER), with CNNs traditionally leading the field. More recently, ViTs have demonstrated strong performance in a variety of image-based tasks by capturing global dependencies and offering flexibility in handling input structures. These architectures show promise for improving emotion recognition accuracy, particularly when dealing with thermal and multimodal data. However, further research is required to determine their effectiveness in low-resolution, low-resource environments typical of IoT deployments.

This research is motivated by the need to bridge the gap between state-of-the-art DL methods and the practical requirements of real-world, embedded emotion recognition systems. It aims to evaluate whether accessible hardware, such as low-cost thermal cameras, can be effectively paired with modern DL architectures to build robust, scalable, and ethical emotion classification models. Furthermore, it seeks to explore the benefits of multimodal fusion, not only to improve accuracy, but also to enhance model resilience in unconstrained and diverse real-world settings.

1.2 Research Objectives

The overarching aim of this research is to explore how low-cost, IoT-based systems can be leveraged for accurate and scalable emotion classification. The work focuses on utilising thermal and visual modalities, combined with state-of-the-art DL architectures, to address current limitations in emotion recognition - namely, the high cost of hardware, limited generalisability of models, and lack of diversity in existing datasets.

In support of this aim, the following specific research objectives have been defined:

1. To advance emotion recognition methods through comprehensive evaluation of unimodal models using established thermal, speech, and text datasets, identifying the strengths and limitations of each modality.

2. To develop a structured and reproducible framework for multimodal emotion data collection, enabling the integration of thermal imaging, speech, and textual data to support robust emotion recognition systems.

3. To design and implement a resource-efficient MER system that fuses thermal, acoustic, and linguistic signals using advanced fusion strategies, and evaluate its feasibility for real-time deployment on edge devices.

To contextualise these objectives in the scope of emotion recognition, the research seeks to investigate whether high-performing models, CNNs and ViTs, can operate effectively on thermal and multimodal data collected using affordable sensors. The study also examines how models trained on one dataset generalise to others and evaluates the potential of hybrid training strategies to improve robustness across varied conditions.

By aligning technical advancements with practical deployment constraints, this research aims to contribute meaningful progress in the development of low-cost, privacy-conscious, and adaptable emotion recognition systems for real-world use.

1.3 PhD Journey

My PhD journey began after completing an MSc in Advanced Computer Systems and gaining experience as a research assistant. This early exposure to academic research solidified my interest in contributing to the field of artificial intelligence. The PhD initially formed part of a collaborative partnership with an industry stakeholder, focusing on the integration of AI technologies into a virtual learning platform. The scope at the time spanned a range of intelligent functions, including automated assessment, emotion recognition, and virtual assistant design. The opportunity to work closely with industry was a strong motivator, as it provided a platform to explore research that had both theoretical depth and real-world relevance.

However, as the collaboration came to an unexpected end, the PhD project continued under the sole supervision of the university. This transition marked a significant turning point in my research focus. I redirected my efforts exclusively toward emotion recognition using IoT technologies - a topic that remained within the broader scope of affective computing but offered a more refined and technically challenging path. The shift allowed me to explore low-cost, deployable solutions using real-time multimodal inputs and cutting-edge DL methods.

Over the course of the project, I published a range of research outputs that reflect the evolution and breadth of the work. These include a paper on text-based emotion classification, a conference paper on real-time emotion recognition, a journal submission detailing the creation and analysis of a novel thermal facial expression dataset, and a journal submission focusing on real-time MER. Each of these studies contributed to building a robust, scalable framework for emotion recognition using low-cost, embedded systems.

The project was not without its technical challenges. Preprocessing complex and varied data, managing DL library dependencies, and designing neural network architectures that balanced efficiency and accuracy were recurring issues that required both persistence and creativity to resolve. Milestones that marked significant progress included the acceptance of the text emotion classification paper, presenting my real-time emotion recognition work at a conference in Trondheim, Norway, completing a bespoke thermal dataset collection, and successfully fusing multiple modalities for a comprehensive journal submission on real-time MER.

Overall, the PhD journey has not only deepened my technical expertise but also strengthened my adaptability and problem-solving skills. It has reinforced the value of applied, practical research in the AI domain and shaped my ambition to continue contributing to intelligent systems that are both innovative and accessible.

1.4 Organisation of the Thesis

This thesis is structured as a compilation of peer-reviewed research outputs, supported by contextual chapters that frame the motivation, methodology, and contributions of each study. The work is organised as follows: Chapter 2 introduces the research, outlining the background, motivation, objectives, and the journey undertaken throughout the PhD. Chapter 3 presents a comprehensive review of related work in textual emotion recognition, with a particular focus on the role of IoT technologies and the utilisation of an edge architecture. Chapter 4 details the methodology adopted for collecting a thermal FER dataset and provides an analysis that includes comparisons with other state-of-the-art datasets using DL models. Chapter 5 integrates the modalities explored in the previous chapters and introduces the speech modality to deliver a comprehensive multimodal analysis in a real-time edge computing setting. The thesis concludes in Chapter 6 with a summary of key findings and outlines potential directions for future research.

Each chapter is designed to be read both independently and as part of a cohesive narrative, collectively contributing to the central aim of developing practical, accurate, and deployable emotion recognition systems using IoT technologies.

Chapter 2

Literature Review

2.1 Introduction

In this chapter, we review existing work in the field of emotion recognition, focusing on both unimodal and multimodal approaches, as well as contributions towards real-time deployment. We begin by setting the context in Section 2.2, highlighting key developments in Machine Learning (ML) and DL methods. In Section 2.2.1, we discuss the advantages of combining multiple modalities, followed by an examination of recent efforts to enable real-time emotion recognition on edge devices in Section 2.2.2, with particular attention given to emerging trends in thermal-based emotion recognition using low-cost and accessible hardware. We then provide a detailed analysis of unimodal literature in the domains of text (Section 2.3) and facial expressions (Section 2.6), and further examine available thermal datasets in Section 2.6 before concluding the chapter in Section 2.7.

2.2 Emotion Recognition

Emotion recognition has long been a focus of affective computing, with early approaches relying on Red Green Blue (RGB) images and handcrafted features to classify emotions [1,2]. Over the past decade, advances in Computer Vision and DL have greatly improved facial emotion recognition accuracy by automatically learning expressive features from large face datasets [3, 4]. Similarly, speech-based emotion recognition has progressed from using prosodic and spectral features to employing deep neural networks on speech spectrograms

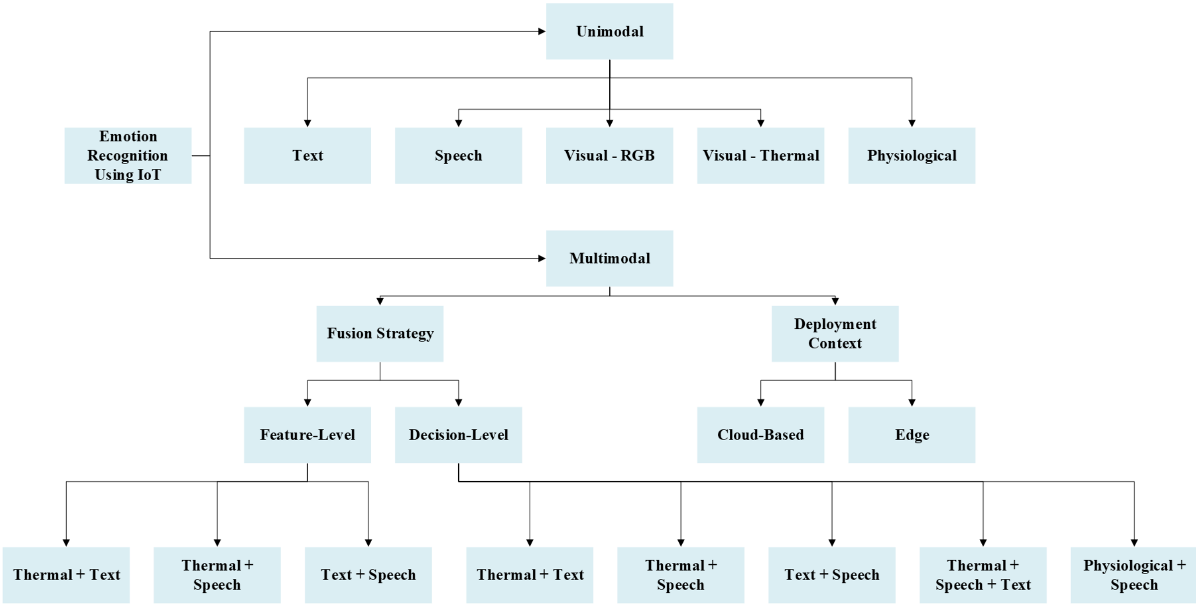


Figure 2.1: Structured taxonomy of emotion recognition approaches

or waveforms and achieving robust performance [5,6]. In the text domain, techniques from sentiment analysis and Natural Language Processing (NLP) are used to detect emotion from written or transcribed speech, by modelling linguistic cues (keywords, syntax, embeddings) that correlate with affective states [7,8]. Physiological signals have also been explored as unimodal inputs for emotion detection in settings where external expressions are limited [9–11].

As the field has expanded to include diverse modalities, and as real-time applications on edge devices have gained traction, a clear categorisation of existing work becomes essential. To support this, Figure 2.1 presents a unified taxonomy of unimodal and multimodal emotion recognition approaches, highlighting fusion strategies, modality combinations, and deployment contexts relevant to this research. To complement the taxonomy, Tables 2.1 and 2.2 summarise representative methods, strengths and limitations, and their suitability for edge or IoT-based deployment.

Each unimodal approach can achieve high accuracy under controlled conditions; for instance, state-of-the-art facial or thermal image classifiers can exceed 90% accuracy on basic emotion classes [9,26]. However, single-modality systems often struggle with robustness in real-world scenarios featuring poor lighting or occlusions, and speech models may misclassify emotions with neutral tone or in noisy backgrounds [27]. These limitations have motivated the integration of multiple modalities to enhance reliability.

Modality	Common Methods	Strengths	Limitations	IoT/Edge Suitability
Text	BoW, TF-IDF, Transformer models (BERT, DistilBERT) [12, 13]	High semantic richness; flexible features	Computationally intensive for large models [14]	Moderate (TF-IDF + lightweight ML models perform well)
Speech	MFCC, CNN-RNN, prosodic features [5, 15]	Rich paralinguistic cues; effective with context	Sensitive to noise; requires preprocessing	Moderate (requires compression/streamlining)
Visual (RGB)	CNNs (e.g., Xception, LeNet), FER-2013, CK+ datasets [16, 17]	Well-established datasets; expressive facial cues	Privacy concerns; low performance in occlusions or poor lighting [18]	Low (high-res input and privacy limits usage)
Thermal Imaging	CNNs, ViTs, low-cost thermal cameras [19, 20]	Privacy-friendly; robust in low-light	Dataset diversity limited; underused on edge [21]	High (suited for ambient edge capture)
Physiological Signals	EEG, GSR, HRV [10, 22]	Direct insight into emotional state	Invasive or impractical sensors	Low (requires specialised hardware)

Table 2.1: Summary of unimodal emotion recognition approaches and their IoT deployment potential

Fusion Strategy	**Modalities Combined**	**Common Methods**	**Strengths**	**Limitations**
Feature-Level Fusion	Text + Audio + Visual	Early concatenation, deep shared representations [6]	Captures fine-grained interactions between features	Sensitive to modality imbalance and alignment issues
Feature-Level Fusion	Thermal + Text, Thermal + Speech	CNN + TF-IDF, ViT + MFCC [20, 21]	Enables compact representation; suitable for edge deployment	Requires synchronized preprocessing; harder to interpret
Decision-Level Fusion	Text + Audio + Visual	Late fusion, majority voting, ensemble models [23, 24]	More flexible; robust to missing modalities	May lose cross-modal interaction detail
Decision-Level Fusion	Thermal + Speech + Text	Independent classifiers + aggregation [19]	High robustness and modularity; easy to parallelise	Increased latency; dependent on classifier calibration
Hybrid Fusion	Text + Audio + Visual	Combined early and late fusion layers [25]	Balances interaction and robustness	Higher complexity and training overhead

Table 2.2: Summary of multimodal emotion recognition strategies using combinations of modalities and fusion types

2.2.1 Multimodal Emotion Recognition

Combining multiple modalities has proven to be an effective strategy to improve emotion recognition accuracy and robustness [22,27,28]. MER systems leverage the complementary information present in different channels. For example, facial expressions provide visual cues, speech provides vocal tone, and language offers contextual semantics. By fusing these signals, MER systems can capture a more holistic representation of the user's emotional state [29, 30].

Numerous studies have demonstrated that multimodal approaches outperform unimodal ones as each modality can compensate for others' weaknesses [31, 32]. Classic MER research focused on audio-visual fusion, demonstrating that combining facial and vocal features produces higher recognition rates than unimodal classification [6, 23, 28]. For instance, emotion classifiers using both voice and face data have reported substantial gains in accuracy and robustness in noisy or dim environments compared to single-modality models [27].

Beyond audio-visual modalities, textual data such as dialog transcripts or social media posts has been incorporated to recognise emotions that are expressed verbally [15, 30]. In conversational settings, leveraging speech content alongside how it is said and shown leads to more nuanced emotion understanding [15, 33]. Similarly, emerging work explores physiological and thermal facial expressions to identify stress or excitement under challenging lighting conditions [9, 26]. Studies have found that fusing thermal images with RGB video can improve accuracy in detecting certain emotions that manifest as temperature changes [26].

A key consideration in is the fusion strategy used to integrate modalities. Fusion can occur at the feature level, also referred to as early fusion, where raw features from different modalities are combined into a joint representation, or at the decision level also referred to as late fusion, where separate classifiers' outputs are merged based on independent modalities [22, 29].

Feature-level fusion approaches require synchronising and aligning data streams and have the advantage of learning cross-modal feature interactions, but they may suffer if the data contains noise or missing data [9, 29]. Late fusion methods treat each modality independently up to the decision stage then use techniques like voting, weighting, or stacking to produce a final prediction [23, 32]. Since each modality can be weighted or ignored dynamically, late fusion offers more flexibility to handle modality-specific failures and often achieves more robust results [23, 32].

Many recent MER systems employ fusion mechanisms beyond simple concatenation or majority vote. Attention-based fusion is a popular technique where the model learns to assign learned weights to each modality's features based on their relevance to the expressed emotion in context [6, 34]. Other work uses gating networks or modality-specific subnetworks that interact allow the model to learn [28, 35].

Overall, the literature shows a clear trend that thoughtfully integrated multimodal systems can achieve higher accuracy and reliability than single-modality systems [29,31,36]. However, simply adding more modalities does not guarantee better performance as the quality and alignment of data and the fusion method are crucial [18,30]. In some instances, bimodal combinations have matched or even outperformed more complex tri-modal systems when the third modality is weak or introduces noise [15, 37].

2.2.2 Real-Time Multimodal Emotion Recognition

Beyond accuracy, other work addresses the deployment of MER in real-time settings. Interactive applications such as social robots, driver monitoring, or healthcare assistants require MER systems that can infer emotions on the fly with low latency [29, 38]. Achieving real-time performance with multimodal inputs is challenging due to the computational cost of processing multiple high-dimensional data streams simultaneously [25, 37]. Prior studies have explored optimisations to meet timing constraints on hardware and software.

On the algorithm side, researchers have designed lightweight DL models and efficient feature extractors that reduce complexity while preserving accuracy, making them suitable for real-time use [27, 33]. For example, using a smaller CNN for facial analysis or an optimised Recurrent Neural Network for audio processing can cut down processing time significantly [25, 38]. Some works execute model compression techniques or knowledge distillation to streamline multimodal networks to achieve faster inference [37]. Others simplify the fusion method by using a simple weighted average or linear classifier in late fusion to minimise overhead at runtime [23].

On the hardware side, leveraging GPUs can accelerate model inference and yield up to 30fps and exploit parallel processing of modalities [11, 33]. Another concern for real-time MER is ensuring consistent performance under streaming condition as live systems must handle continuous input and possibly uncertain modality alignment [25, 29]. Recent research addresses these by using sliding windows, enabling the model to output emotion predictions continuously [11].

Some studies have implemented MER on embedded platforms such as service robots or smartphones [29, 33, 39]. These instances demonstrate multimodal models can run on edge devices and still achieve real-time inference, making them practical for ubiquitous applications [33, 38]. However, maintaining high accuracy with low latency often involves trade-offs as extremely lightweight models may sacrifice some recognition performance and high frame rates can be sustained only by limiting input resolution or frequency, therefore, recent works strive to balance the accuracy-speed trade-off [25, 37].

The literature confirms it's possible that all the tested multimodal configurations can be deployed with high throughput given appropriate optimisations [27, 33]. System-level evaluations in prior work report that these models can meet real-time requirements while still significantly outperforming their unimodal counterparts in accuracy [25, 36, 38].

In summary, prior research provides a foundation for our work by illustrating the strengths of various modalities for emotion recognition, the benefits of fusing multiple modalities, and the practical considerations for deploying such systems in real time. Our approach builds on these insights by specifically examining thermal imagery alongside speech and text, applying both early and late fusion, and evaluating not only accuracy but also the runtime performance to ensure suitability for edge deployment.

2.3 Textual Emotion Recognition

In textual emotion recognition recent research has increasingly focused on the analysis of user-generated content from platforms such as social networks, blogs, instant messaging services, and other multimedia sources. This analysis has become pivotal for understanding individuals' sentiments and emotions, employing a variety of established techniques, including Bag of Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF), as well as more contemporary methods such as embeddings and DL models. [40]

proposed a DL framework for fine-grained emotion detection, utilising a corpus derived from a television showâĂŹs transcript. Their methodology incorporates a sequence-based CNN, enhanced through word embeddings and an attention mechanism. While such approaches have demonstrated high effectiveness, they are often computationally demanding, thereby limiting their suitability for deployment on resource-constrained edge devices.

Similarly, [41] introduced the Semantic-Emotion Neural Network (SENN) which combines semantic and syntactic information from emotional content through word embeddings. Their work integrates a BiLSTM for semantic context and a CNN for feature extraction, producing three model variants utilising Word2Vec, GloVe, and FastText embeddings. FastText embeddings showed superior performance, emphasising the importance of embedding selection in emotion classification. While these models show high accuracy, they generally require significant computational resources, posing challenges for real-time applications on edge devices.

Unlike the use of BiLSTM, [42] developed the Sentiment and Semantic Based Emotion Detector (SS-BED), a DL model based on an architecture comprising LSTM networks that integrates sentiment and semantic information using two separate word embedding matrices. This approach encodes sequential patterns in user utterances, making it a valuable contribution to emotion detection from textual data. However, this approach also relives heavily on intensive computational elements, thus limiting its scalability in environments low on computational resources.

Additionally, [43] explored concerns raised by migrant workers, sourced from a microblogging site (Weibo), where key issues like wages, education, and healthcare were discussed. Their model, combining Word2Vec and TF-IDF features within a DL framework (BI-LSTM and CNN), highlighted key areas requiring government attention.

Transformer models have also made significant advances in emotion recognition. [44] compared Bidirectional Encoder Representations from Transformers (BERT), RoBERTa, Distillated-Bidirectional Encoder Representations from Transformers (DistilBERT), and XLNet using the ISEAR dataset, [45], which includes emotions such as anger, disgust, sadness, fear, joy, shame, and guilt. Their findings demonstrate that RoBERTa achieved the highest accuracy with superior computational efficiency. Despite their state-of-the-art performance, models like BERT, RoBERTA and XLNet typically demand high computational power and memory, making them impractical for deployment on edge devices, with the authors highlighting potential inefficiencies when scaled down to smaller platforms.

Expanding on transformer models, [46] proposed Emotion-Sentence-DistilBERT (ESD-BERT), utilising a Siamese Network-based Sentence-BERT to reduce complexity through Knowledge Distillation. ESDBERT, using two DistilBERT models for feature extraction and pooling methods for feature crossing, outperformed BERT and DistilBERT models with 50% faster processing, demonstrating that transformer-based approaches can be both accurate and computationally efficient.

Building upon the advancements in transformer architectures, the exploration of Generative Pre-trained Transformer (GPT) models has garnered considerable interest. [47] examined GPT embeddings for emotion classification which highlighted the ability of GPT embeddings to encode rich linguistic data, thus improving the accuracy and effectiveness of ML models for emotion detection. Additionally, the comparative studies by [48] between GPT models and BERT models for emotion recognition shed light on the adaptability and versatility of GPT in contrast to task-specific models like BERT.

The general-purpose nature of GPT models presents promising flexibility, especially highlighted by the unexpected competitive performance of GPT-3.5 in [49] despite its lack of explicit task-specific training. These studies underscore the transformative potential of leveraging large language models across diverse NLP applications. The exploration of prompt design and evaluation frameworks becomes crucial for harnessing the full capability of large language models in emotion classification.

Traditional ML approaches have also been employed for emotion recognition. [50] used an ALBERT-BiLSTM model combined with Support Vector Machines (SVM)-NB classification. This approach utilises the ALBERT model for text preprocessing and BiLSTM for feature extraction, outperforming other general models. [51] proposed a Voting Classifier (VC) based on TF-IDF feature representations, achieving an accuracy of 79% using TF-IDF representations, validated across binary and multiclass datasets. This highlights the continued relevance of traditional classifiers in specific contexts.

Additionally, [52] introduced a system for fine-grained emotion detection using an ensemble classifier consisting of KNN, Multilayer Perceptron, and DT, enhanced with Tree-structured Parzen Estimator (TPE) for hyperparameter tuning, demonstrating high accuracy in detecting fine-grained emotions in both regular and irregular sentences.

Twitter data remains a common source for emotion classification. [53] developed Emotex and EmotexStream, systems for automatic emotion classification in tweets, with real-time classification enabled by EmotexStream. Similarly, [54] proposed an approach that integrates message replies, incorporating parameters like emotion scores, sentiment, and agreement, providing new insights into user behaviour on social platforms.

Recent research explores many approaches to improve efficiency. [55] utilised features from adjacent sentences and modified the KNN algorithm to allow iterative corrections. [56] proposed a Deep Neural Network with Leaky ReLU activation, including a step after pre-processing where features are ranked prior to classification. [57] adopted a DL approach, incorporating slang word and repetitive word replacements during preprocessing, and demonstrated that the Light Gradient Boosting Machine was the most efficient model compared to other popular DL methods.

In summary, the textual emotion recognition literature shows a strong trend towards DL and transformer-based models for emotion recognition, which often outperform traditional classifiers in terms of accuracy. However, these models typically demand significant computational resources, limiting their practical application on edge devices. This work addresses a critical gap by exploring traditional approaches with a focus on computational efficiency. Our research is distinct in its application of models such as Stochastic Gradient Descent (SGD) and SVM in resource-constrained environments, thus enabling deployment on low-computation devices like edge boards. This contributes to more practical implementations of emotion classification in real-world settings where computational resources are limited.

2.4 Facial Expression Recognition

While textual emotion recognition offers valuable insights, it is often limited by the absence of non-verbal cues which are critical for accurately interpreting emotional states. As a result, research attention has increasingly turned towards visual modalities, particularly facial expressions, which convey rich emotional information.

FER has been an active area of research due to the rise of CNNs and their ability to process data effectively. A wide variety of approaches have been proposed in recent years using visible spectrum cameras leveraging established datasets such as FER-2013 [58], KDEF [59], JAFFE [60], and CK+ [61]. Regardless of thermal of visible spectrum each approach follows the same process of removing backgrounds in images, detecting the face and classifying emotions based on further processing techniques.

In [16] the authors employed the FER-2013 dataset and a CNN model to predict emotions from facial expressions using images captured by visible spectrum cameras. Similarly, [62] introduced a custom dataset with visible spectrum images images and applied CNNs to classify emotions. This approach was complemented by [63] who used the same dataset and utilised the XCeption CNN architecture [17] to build a real-time system.

The authors in [64] expanded upon the Xception model's utility and using the same dataset deployed real-time emotion recognition on a Raspberry Pi, detecting subtle facial micro-expressions to classify emotions. Similar to this work they showcase the feasibility of deploying a real-time system on resource-constrained hardware.

Other approaches have similarly focused on real-time processing. For instance, [65] used a CNN and KDEF dataset where they also executed face recognition as well as the emotion. Work in [66] propods a real-time CNN-based system which merged KDEF, JAFFE and their own custom dataset and employed the LeNet architecture to conduct real-time analysis at 30 video frames per second.

A unique architecture was proposed in [67] who introduced an edge-cloud-based system which captured facial images and speech signals. Processing data on the edge, they utilised a secret-sharing scheme to distribute data across edge clouds while the core cloud handled feature extraction using CNNs.

Beyond the visible spectrum thermal imaging has emerged as an alternative approach for emotion recognition, particularly in cases where privacy concerns are paramount. In [20], they utilise thermal cameras to detect emotional cues based on heat distribution across the face by training YOLO [68] using the Natural Visible and Infrared Facial Expression Database (NVIE) dataset [69]. The authors in [70] also used a transfer learning approach with the NVIE dataset by combining AlexNet [71] and a Support Vector Machines classifier.

Other approaches have incorporated different techniques, such as [72]. They divided facial images into regions to focus only on the eyes and lips to classify emotions. The authors in [73] also proposed a region-based approach which focuses on the eyes and nose and utilise a multiple-instance learning algorithm to track facial features in video files.

2.5 Thermal Facial Expression Recognition

Due to advancements in thermal datasets, there has been a significant impact on the performance of facial expression recognition models that utilise different architectures and preprocessing approaches outwith the traditional visible hardware. The authors in [74] introduced the InfraRed Facial Expression Network (IRFacExNet), a DL model that capitalises on thermal imaging features, such as low-light conditions where visible spectrum cameras struggle. Their model outperformed conventional methods. Similarly focusing on low-light conditions, work in [75] utilises a CNN architecture and bases their analysis on four regions of the face. While their results were promising, the study's limited focus on only four facial regions may overlook the complexity of emotional expressions that involve broader facial movements. In [76], the authors present TIRFaceNet, with an architecture tailored for robustness against varying environmental conditions, and report admirable accuracy on both the DHU and DHUFO datasets. However, to the best of our knowledge, there are no insights into the model's limitations, and limited information or no reference is given for the dataset.

The availability of multimodal datasets has enabled researchers to conduct comparative analyses between thermal and visible spectrum images. Using the Gray Level Cooccurrence Matrix (GLCM) method to extract statistical features, [77] employed a custom CNN alongside SVM classifiers. Their findings indicate that thermal images outperformed visible images by achieving higher accuracy, although the proposed approach's reliance on statistical features may not fully capture the nuances of emotional expressions. Unlike statistical features, the authors in [78] proposed an architecture based on a modified version of ResNet152 to enhance feature extraction capabilities. Their study reported metrics; however, due to the complexity of the architecture, challenges may arise in terms of computational efficiency and applications operating in real-time.

Many popular models based on the CNN architecture exist, and often approaches use the base model and conduct transfer learning. [79] present TERNet - an architecture adopting the VGG-Face CNN model. Their findings highlight the potential of transfer learning in enhancing facial expression recognition accuracy while bypassing the need to train complex models from scratch. The authors in [80] conducted a comprehensive analysis of feature-based facial expression recognition, which provided valuable insights into feature extraction techniques both locally and globally. Contrary to [79], their findings reported that features extracted using the VGG CNN network performed poorest on the Kotani Thermal Facial Expression (KTFE) and NVIE datasets. This suggests that different models and approaches are better suited to different datasets. Also adopting a transfer learning approach, the authors in [81] utilise AlexNet for feature extraction and classification. They enhanced the architecture by creating a hybrid model that combined AlexNet and SVM to yield the best results, with thermal images' accuracy outperforming visible images.

The reviewed work highlights the many approaches that can be taken utilising thermal images, ML classifiers, DL models, and transfer learning, which show promise in improving recognition accuracy. However, while DL models have shown significant improvements, issues related to overfitting, computational efficiency, and real-time applicability remain under-explored. Furthermore, existing work tends to use datasets where data has been collected using expensive and often commercial thermal cameras. In this work, we intend to demonstrate model performance using accessible hardware combined with DL techniques.

2.6 Thermal Facial Expression Datasets

Thermal camera datasets have become increasingly significant in facial expression recognition. Although not as prevalent as visible spectrum datasets, thermal datasets serve as crucial resources for training and testing algorithms operating under varying conditions, poses, and expressions. A summary of prominent thermal camera datasets can be seen in Table 2.3.

A popular dataset among researchers is the KTFE dataset introduced by [82]. It focuses on various facial expressions; however, the demographic is limited to Vietnamese, Japanese, or Thai nationals, which could adversely affect models trained using this dataset. The same authors also released KTFE v2 [83], this time integrating different modalities for a more comprehensive analysis. The same issues exist in this dataset, as the same range of nationalities was used.

Another popular dataset is NVIE, introduced by [84], which incorporates both visible and thermal spectrum images. This dataset is a comprehensive collection with different variations in angles of poses, glasses on and off, and various expressions. However, the dataset's complexity may pose challenges in terms of processing, and it is stored on a Chinese data storage platform, making it difficult for non-Chinese speakers to access. Other notable mentions of popular datasets include IRIS [85] and the NIST/Equinox database by and Equinox Corporation (http://www.equinoxsensors.com/products/HID.html).

The Tufts Face Database, as described by [86], offers a comprehensive collection of images across multiple modalities, which include images in the thermal spectrum. This dataset encompasses over 10,000 images from a diverse group of individuals, and although the inclusion of other modalities allows for cross-comparison and benchmarking of algorithms, the dataset's complexity may pose challenges in terms of processing and analysis.

The authors in [87] introduced a dataset containing various posed expressions across different angles under controlled conditions. A notable weakness of their dataset is the limited diversity, as it primarily features a narrow demographic, which may affect the generalisability of algorithms developed using this data. [88] also collected their data under controlled conditions and conducted a comparative analysis between thermal and visual modalities. This data may be limited, as environmental factors could influence the results of future inference by models trained using it, which may not be accounted for in all scenarios.

[19] introduced a thermal dataset including rigorous annotation of facial expressions in their approach. The comprehensive nature of the annotations allows for detailed analysis of facial expressions. However, similar to [87], the demographic representation may be limited, as there is no mention of it, potentially impacting the performance of models in real-world applications where diversity is crucial.

Dataset	Modality	Expressions Included	Demographic Diversity	Elicitation Method	Limitations
KTFE [82]	Thermal	anger, disgust, fear, happy, sad, surprise, neutral	Low	Spontaneous	Limited demographic diversity; restricted to specific nationalities (Vietnamese, Japanese, Thai)
KTFE v2 [83]	Multimodal	anger, disgust, fear, happy, sad, surprise, neutral	Low	Spontaneous	Same demographic limitations as KTFE (Vietnamese, Japanese, Thai); limited generalisability
NVIE [84]	Multimodal	anger, disgust, fear, happy, sad, surprise	Moderate	Posed + Spontaneous	Complex data structure and preprocessing; hosted on a Chinese platform, making access difficult for non-Chinese users
IRIS [85]	Multimodal	anger, laughing, surprise	Moderate	Posed	Lacks detailed documentation on expressions, demographics, and elicitation methods

Dataset	Modality	Expressions Included	Demographic Diversity	Elicitation Method	Limitations
NIST Equinox	Thermal	smiling, frowning, surprised	Unknown	Posed	Limited public documentation; commercial dataset with restricted access; unclear participant demographics and methodology
Tufts [86]	Multimodal	neutral, smile, closed eyes, shocked	High	Posed	Complex multi-modal structure not tailored specifically for facial expression recognition; thermal data can be difficult to isolate; higher processing overhead
Kowalski et al. [87]	Thermal	smiling, sad, surprise, angry	Limited	Posed	Demographic details such as age, gender, and ethnicity not reported; limited subject diversity
Wesley et al. [88]	Multimodal	surprise, fear, sadness, disgust, anger, happiness	Moderate	Posed	Small sample size; limited scalability to real-world environments with natural variability

Dataset	Modality	Expressions Included	Demographic Diversity	Elicitation Method	Limitations
Kopaczka et al. [19]	Thermal	neutral, happiness, sadness, surprise	Limited	Posed	Small participant pool with limited demographic diversity; demographic details (age, gender, ethnicity) not specified; expressions limited to a subset of basic emotions
Ours	Thermal	anger, fear, happy, neutral, sad, surprise	Moderate	Posed + Spontaneous	Small sample size; lacks in-the-wild data

Table 2.3: A comparison of thermal facial expression datasets

This literature highlights significant advancements in the development of thermal facial expression databases. While these studies provide valuable data and methodologies, they also share common weaknesses such as limited demographic diversity and environmental influences. Furthermore, these studies use expensive cameras often unavailable to general consumers to collect their data, which is a gap in the literature this work intends to fill.

2.7 Conclusion

In summary, the reviewed literature demonstrates significant advancements in the field of emotion recognition, from early unimodal approaches to sophisticated multimodal and real-time systems. Unimodal methods, while effective under controlled conditions, often suffer from reduced robustness when applied to dynamic and complex real-world scenarios. This has driven considerable research interest towards multimodal fusion strategies, where the complementary strengths of visual, auditory, and textual modalities are leveraged to improve classification performance and system reliability.

Furthermore, with the increasing demand for real-time affective computing applications, recent studies have explored efficient model architectures, fusion techniques, and hardware optimisations to meet the constraints of edge deployments. Thermal imaging, in particular, has emerged as a promising modality for privacy-preserving and low-light emotion recognition, although challenges remain in terms of dataset diversity and model generalisability.

Building on these insights, this thesis addresses key gaps in the literature across the following chapters. Chapter 3 explores computationally efficient textual emotion classification techniques suitable for deployment on low-cost edge devices, providing a foundation for unimodal analysis. Chapter 4 investigates the feasibility of using low-cost thermal imaging for facial emotion recognition, introducing a novel dataset and comparing CNN and ViT architectures to assess model generalisation. Chapter 5 integrates thermal, speech, and text modalities using multimodal fusion strategies and evaluates the performance of real-time emotion recognition systems on edge platforms. Collectively, these chapters contribute toward advancing the practical deployment of robust, scalable, and accessible emotion recognition systems in resource-constrained environments.

Chapter 3

Textual Emotion Recognition

3.1 Introduction

Text classification is a process in the field of NLP which encompasses a diverse array of specific tasks, including but not limited to sentiment analysis, question classification, and topic classification. It presents a perennial challenge: the selection of optimal models and feature representations that are best suited for each specific task. This chapter aims to delve into this challenge by undertaking a comparative analysis of the performance of nine distinct ML classification models and a DistilBERT model, utilising three different feature representations.

Text classification can be executed through various approaches, which range from rule-based systems, [89], through traditional ML, [90], to more contemporary DL methods, [91]. Each of these methods has its own merits, offering varying degrees of success. However, DL models are often computationally intensive, making them less suitable for edge computing environments where resources are limited. In this work we show that traditional ML methods, with their lighter computational footprint, provide a feasible alternative for emotion classification on edge devices with often comparable or improved accuracy. A pivotal factor influencing the performance of ML models is feature representation, with traditional methods such as BoW, [92], and TF-IDF, [93], placed against more modern techniques such as word embeddings, [94] or transformer models like BERT, [12].

Emotion classification, a subset of text classification, involves the identification of fine-grained emotions such as happiness, sadness, and anger from textual sources, [14]. These sources often take informal forms such as social media posts, text messages, or blog content, a trend that is increasingly prevalent in the field.

Edge computing has many use cases for text classification. For example, [95] proposed a sentiment analysis system for an android mobile, [96] proposed a topic sentiment analysis system which formed as part of an edge social system, and [97] proposed an edge system to classify news articles. Although a great deal of emotion classification studies featuring edge systems focus on classification by images, adopting a multi-modal approach such as work by [98] enables speech transcriptions to be considered which form part of the edge architecture. The inefficiencies of deploying DL models on edge devices underscore the motivation for this study [99], highlighting the potential for significant improvements by employing traditional ML methods.

This chapter considers edge devices and utilises a dataset curated by [100], which is composed of tweets categorised into six distinct emotions: surprise, love, fear, anger, sadness, and joy. The dataset serves as a foundation for evaluating the effectiveness of each classification model and feature representation in the context of emotion recognition.

Our analysis provides an exhaustive examination of various models including SVM, Logistic Regression (LR), Random Forest (RF), SGD, Decision Tree (DT), Complement Naive Bayes (CNB), Gaussian Naive Bayes (GNB), Multinomial Naive Bayes (MNB), K-Nearest Neighbors (KNN), and DistilBERT. These models are evaluated using BoW, TF-IDF, and word embedding features to assess their performance and overall efficacy.

The investigation employs an edge architecture approach, where emotion classification can take place using an edge board. Model training and inference times are considered, aiming to identify which combination of model and representation offers the most efficient solution in terms of both performance and training time. The main contributions of our work include:

- An exhaustive evaluation of nine ML classification models' performance on a dataset of tweets categorised into six emotions.
- The optimal hyperparameters for each ML classifier.
- An analysis of the most efficient feature representation and model combination, particularly suitable for scenarios involving low computational devices.

- Each model and feature representation combination latency using an edge board

The structure of this chapter is as follows: Section 3.2 delineates our research methodology, encompassing dataset preprocessing, feature set extraction and ML classification models. Section 3.3 details the results derived from each feature representation, discussing each set of outcomes. Finally, Section 3.4 concludes the chapter, summarising our findings and contributions.

3.2 Methodology

The research method outlines the various stages involved from a macro level perspective, comparing the performance of each feature set and ML classifier. Preprocessing consisted of cleansing the dataset and undertaking a series of steps detailed later in this section to prepare the data for the feature extraction phase. Feature extraction involved performing calculations on the processed data to generate BoW, TF-IDF, and word embedding features, which were then fit to each respective ML classifier in the model training phase. In the model hypertuning phase, each ML classifier was fit with a range of hyperparameters to identify the most optimal configuration of the classifier. Finally, the model evaluation phase used the testing data to infer each model and employed various metrics detailed later in this section to evaluate each model's classification performance and inference performance.

3.2.1 Dataset

This study utilises the CARER emotion dataset collated by [100], comprising English tweets categorised into six emotions: anger, fear, joy, love, sadness, and surprise. The dataset's initial collection involved the use of noisy labels, which were later refined through distant supervision as described by [101]. For this analysis, the dataset consists of 18,000 training samples and 2,000 test samples.

The length of each tweet varied from 2 to 66 words, though the overall pattern in word count remained relatively consistent. The dataset distribution reveals the surprise class consists of 652 training samples and 66 test samples; the love class includes 1,482 training samples and 159 test samples; the fear class contains 2,149 training samples and 224 test samples; the anger class comprises 2,434 training samples and 275 test samples; the sadness class has 5,216 training samples and 581 test samples; and the joy class possesses 6,066 training samples and 695 test samples. The imbalance in the dataset is evident, with the surprise and love classes having significantly fewer instances compared to sadness and joy.

The authors in [102] demonstrate that BoW and TF-IDF features are suitable for handling dataset imbalance, and their findings align with our results in Section 3.3, particularly in relation to neural networks. Some studies, such as [103], indicate that oversampling techniques can negatively impact performance, where models like KNN and LR showed worse results. Additionally, [104] found that certain models achieved their highest evaluation metrics when datasets were imbalanced, rather than when they were balanced. Addressing the imbalance of the dataset is in the scope for future work.

3.2.2 Cleansing and Preprocessing

The text in the dataset was preprocessed before training the classification models. Sentences were tokenised into individual words using the Natural Language Toolkit (NLTK) Python package [105], ensuring a detailed level of analysis. Tokens were standardised to lowercase to maintain case-insensitivity during feature extraction, and non-essential symbols and punctuation were removed to focus on linguistic content. Common stop words, which carry minimal semantic value, were eliminated using NLTK's stop word library. Words were then lemmatised, reducing them to their base forms while considering context and morphology, which helped lower dataset dimensionality and improve model generalisation. Table 3.1 illustrates five randomly selected extracts from the dataset, demonstrating both the raw textual entries and their corresponding preprocessed formats.

Following preprocessing, the text was then primed for the feature extraction process using the selected representation methods, before being fitted to the ML classification models.

3.2.3 Feature Extraction Techniques

Prior to the application of classification models, it is imperative to transform the preprocessed textual data into a format that is comprehensible to the classification models. This

Text State	**Textual Representation**
Raw text	i feel pretty pathetic most of the time
Preprocessed	feel, pretty, pathetic, time
Raw text	ive been feeling a little burdened lately wasnt sure why that was
Preprocessed	ive, feeling, little, burdened, lately, wasnt, sure
Raw text	i am ever feeling nostalgic about the fireplace i will know that it is still on the property
Preprocessed	ever, feeling, nostalgic, fireplace, know, still, property
Raw text	i now feel compromised and skeptical of the value of every unit of work i put in
Preprocessed	feel, compromised, skeptical, value, every, unit, work, put
Raw text	i feel like i have to make the suffering i m seeing mean something
Preprocessed	feel, like, make, suffering, seeing, mean, something

Table 3.1: Five random extracts from the CARER dataset

transformation is achieved through feature extraction, a process that converts text into numerical representations which encapsulate the essential information inherent in the text. Different feature extraction methods are tailored to capture various aspects of linguistic and syntactical information, each contributing uniquely to the overall effectiveness of the classification process.

Bag of Words

The BoW model is a prevalent technique in NLP for text modelling and feature engineering. This method transforms text data, referred to as documents, into vectors. BoW primarily focuses on the occurrence of words, creating a feature represented by a fixed-length vector composed of 1's and 0's. While other adaptations of BoW exist that track word counts instead of using a binary representation, this study employs the binary approach.

The simplicity of the BoW approach is one of its primary advantages, providing a straightforward method for ML algorithms to learn from textual data. The calculations required for BoW are generally inexpensive to compute as they do not necessitate extensive analysis to extract semantic or structural information from the text documents.

Term Frequency-Inverse Document Frequency

TF-IDF is an advanced feature extraction technique used to evaluate the significance of terms within a collection of documents. It operates on the principle that words appearing frequently in a specific document but relatively rarely across the entire training corpus are likely to be more indicative of that document's content.

The Term Frequency (TF) component of TF-IDF quantifies the frequency of a term within a specific document. It is computed by dividing the number of occurrences of the term by the total number of words in the document. This measure provides a raw count of the term frequency, reflecting how often a term appears in the document, which is then normalised against the length of the document.

The Inverse Document Frequency (IDF) component assesses the rarity or commonality of a term across the entire training corpus. It is calculated by taking the logarithm of the ratio of the total number of documents to the number of documents containing the term. The logarithmic scale is used to dampen the effect of terms that appear in many documents, thereby ensuring that common words like 'the' or 'is', which appear in many documents, don't overshadow the importance of more unique terms that could be more relevant to the specific content of a document.

The combination of TF and IDF scores in TF-IDF facilitates the balancing of the frequency of terms within individual documents against their frequency across all documents, thereby providing a more nuanced and effective way of extracting features from the textual data.

DistilBERT Embeddings

Word embeddings represent words in a numerical format and offer a significant advancement over traditional feature extraction techniques like BoW and TF-IDF. Unlike these earlier methods, word embeddings are adept at capturing the semantic relationships between words through analysing large amounts of text data and identifying patterns in word usage. DistilBERT uses a multi-layer bidirectional transformer encoder to generate contextualised representations of word in a sentence and compresses the knowledge of BERT into a smaller, more efficient model [12].

3.2.4 Machine Learning Classifiers

In this work we include nine different ML classifiers combined with the feature representations detailed previously. The algorithms that were included and assessed in this study include: Support Vector Machine, Logistic Regression, Random Forest, Stochastic Gradient Descent, Decision Tree, Complement Naive Bayes, Gaussian Naive Bayes, Multinomial Naive Bayes, and K-Nearest Neighbors. These algorithms are commonly used for text classification tasks [106, 107] and have been proven to perform well [108–110].

3.2.5 Machine Learning Hyperparameter Tuning

In order to determine the optimal hyperparameters for each ML model, the GridSearchCV module from the Scikit-Learn Python package, [111], was employed. GridSearchCV systematically explores a range of specified parameters, fitting the data to the ML model and iterating through each combination of parameters. To evaluate accuracy, cross-validation is utilised, with this study opting for a five-fold cross-validation approach within each respective test set. The specific range of parameters explored during the hyperparameter tuning of each model is detailed in Table 3.2. Due to the computation power required, the tuning phase was not done using the edge board.

3.2.6 Model Evaluation

The performance of the ML classification models in this study was assessed using a suite of metrics: accuracy, precision, recall, and F1 score. Each of these metrics offers a distinct perspective on the effectiveness of the models and are calculated based on the predicted True Positives, False Positives, and False Negatives. Latency was also considered which measures the inference time.

These metrics collectively enable a comprehensive evaluation of each classification model's performance and efficiency. They provide a holistic understanding of how well each model handles the nuances within the various feature representations of the textual data and what effect this has when we are faced with low computation challenges.

Model	Search Parameters
SGD	loss: hinge, log, squared_hinge, modified_huber; alpha: 0.0001, 0.001, 0.01, 0.1; penalty: l2, l1, none
SVM	C: 0.1, 1, 10, 100; gamma: 1, 0.1, 0.01, 0.001; kernel: rbf, poly, sigmoid
LR	penalty: l1, l2; C: np.logspace(-3, 3, 7); solver: newton-cg, lbfgs, liblinear
RF	n_estimators: 200, 500; max_features: auto, sqrt, log2; max_depth: 4, 5, 6, 7, 8; criterion: gini, entropy
DT	max_leaf_nodes: range(2, 100); min_samples_split: 2, 3, 4
CNB	alpha: 0.01, 0.1, 0.5, 1.0, 10.0; fit_prior: True, False; norm: True, False; class_prior: None, 0.1 * len(classes)
GNB	var_smoothing: logspace(0, -9, num=100)
MNB	alpha: 0.01, 0.1, 0.5, 1.0, 10.0; fit_prior: True, False; class_prior: None, 0.1 * len(classes)
KNN	n_neighbors: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Table 3.2: Machine learning classifiers and the range of hyperparameters

3.2.7 Tools and Libraries Used

In transforming the training data into each respective feature representation vector, a script was developed using Python 3.10 and Visual Studio Code as the primary programming environment. For traditional ML models combined with BoW and TF-IDF features, a Raspberry Pi 4B was used to run the script and trigger the preprocessing, training, and testing phase. In contrast, the preprocessing and training phase for the DistilBERT model took place on an Intel Core i7 processor-based machine running Windows 11 with an Nvidia GeForce RTX 3060Ti graphics processing unit.

The Natural Language Toolkit (NLTK) Python package was crucial in the preprocessing stage, [105], where it played a vital role in tasks such as tokenising the textual data, removing stop words, and performing lemmatisation on each item. This preparation of the data enabled subsequent analysis.

Scikit-Learn's TF-IDF Vectorizer module was utilised to generate Term Frequency-Inverse Document Frequency representations of the data, transforming the corpus into a format amenable to ML analysis. Scikit-Learn's Metrics module was employed to produce

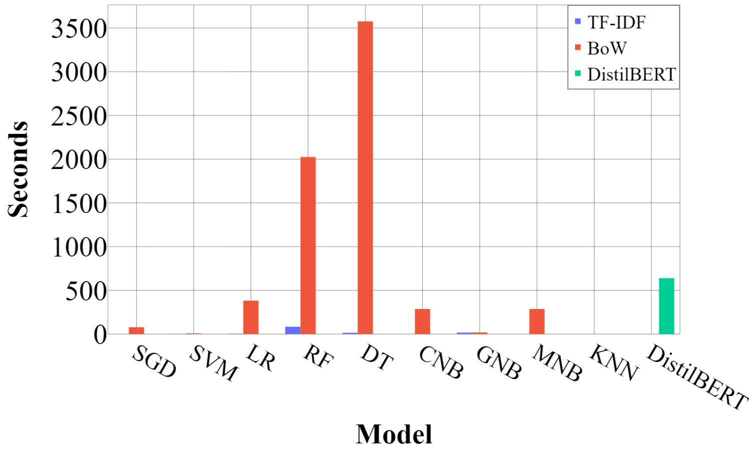


Figure 3.1: Model training times by feature set

comprehensive classification reports, facilitating the calculation of various classification metrics essential for assessing the performance of the models.

3.3 Results and Discussion

This chapter delves into the performance outcomes of the various classification models implemented, with a rigorous examination of each algorithm's capacity to discern and accurately identify emotions embedded within the text. The results not only shed light on the optimal tuning parameters for each model but also explore the dynamic interplay between the classification models and the different feature representations employed. Each ML classifiers performance was evaluated using the default model hyperparameters before being compared with the hyperparameters discovered in the tuning phase. Next, training times and inference times were recorded to give insights in to each models efficiency.

3.3.1 Model Training Times

After completing the data processing and feature extraction phases, each feature set was used to train the nine classification models. The model training times for each respective algorithm are displayed in Figure 3.1, providing valuable insights into the computational performance of each model during the training phase.

The SGD model had a relatively fast training time of 0.68 seconds for TF-IDF features, while the BoW representation took 78.92 seconds. In contrast, the model displayed efficient training times, with the TF-IDF feature set requiring 1.4 seconds and BoW taking 8.42 seconds. The LR model training time took 4.71 seconds for TF-IDF features, while BoW took significantly longer at 382.81 seconds.

The longest training times were observed with the RF and DT models using the BoW feature set, taking 2026.04 seconds and 3576.45 seconds, respectively. The RF model recorded the highest training time of 83.95 seconds with TF-IDF features, while the DT model took 15.1 seconds.

The CNB, MNB, and KNN models had low training times when using TF-IDF features, but with BoW features, they took significantly longer except for KNN which took 0.005 seconds to train. On the other hand, the GNB model performed similarly in both feature sets, taking 17.57 seconds with TF-IDF features and 19.93 seconds with BoW features.

Unlike the other ML models, fine-tuning the DistilBERT model was done using a 3060Ti graphics processing unit due to the memory requirements, and it took a total of 640.2 seconds for 2 epochs. Running the fine-tuning process for 2 epochs produced the optimal configuration before the loss and accuracy became inadequate, as shown in Figure 3.2.

The findings highlight significant variations in training times across different models and feature sets, indicating efficient trade-offs between model complexity and computational loads. The more complex DT and RF models displayed longer training times with the BoW feature set due to its high dimensionality and sparsity. In contrast, models such as SGD, SVM, CNB, MNB, and KNN showed faster fit times with TF-IDF, demonstrating that dense, lower-dimensional feature representations can significantly reduce computational loads.

These findings are not only relevant for understanding the computational demands of various ML models but also offer practical guidance for deploying real-time applications on edge devices, where model scalability and speed are crucial factors. Inference times, which are equally crucial for evaluating model efficiency, are discussed in Section 3.3.5.

3.3.2 Bag of Words Feature Set

The results derived from using the BoW feature representations provide insights into the performance of the classification models, employing a straightforward approach that relies purely on syntactical matches.

The hyperparameter tuning outcomes offer valuable estimations for the the optimal configuration of each model. SGD model was found to perform optimally with a learning rate (alpha) of 0.001, utilising a modified Huber loss function and an L2 penalty. In the case of the SVM model, its best performance was achieved with a regularisation parameter (C) of 0.1, alongside a square hinge loss function and the application of an L2 penalty. The LR model demonstrated increased accuracy when set up with a liblinear solver and an L2 penalty.

For the RF model, the ideal configuration was identified with the use of the Gini impurity criterion, a maximum tree depth of 8 (equivalent to the square root of the number of features considered when searching for the best split), and 200 estimators. The DT model was estimated to perform most effectively when configured with a maximum of 267 leaf nodes.

The CNB model's best performance was calculated with an alpha of 1.0, no class prior, fit prior set to true, and norm set to false. The GNB model was estimated to achieve optimal performance when its variance smoothing parameter was set to approximately 0.0187. The MNB model was predicted to be most effective with an alpha of 0.5, no class prior, and the fit prior set to false. Finally, the KNN algorithm was found to perform best when configured with just 1 neighbor. The result of the models when fitted with the optimised parameters are shown in Table 3.3

The data presented in Table 3.3 shows for the SGD model, hyperparameter tuning resulted in a slight improvement in recall but had a marginal negative impact on accuracy. The SVM model exhibited a modest enhancement in both accuracy and precision. The LR model showed a marginal increase in precision but otherwise remained largely unaffected by hyperparameter tuning. In contrast, hyperparameter tuning proved to be significantly detrimental to the RF model, negatively impacting all evaluated metrics. The DT model, however, displayed slight improvements across all metrics, indicating a positive response to hyperparameter tuning.

Regarding the Naive Bayes classifiers, hyperparameter tuning had minimal effect on the CNB model. However, the GNB model experienced a substantial boost in performance across all metrics, particularly in accuracy and recall. The MNB model also showed improvements in accuracy, recall, and the F1 score. On the other hand, the KNN model

Algorithm	Accuracy	Precision	Recall	F1 Score
SGD baseline	0.90	0.85	0.82	0.84
SGD hypertuned	0.89	0.85	0.83	0.84
SVM baseline	0.89	0.84	0.84	0.84
SVM hypertuned	0.90	0.85	0.84	0.84
LR baseline	0.89	0.84	0.83	0.83
LR hypertuned	0.89	0.85	0.83	0.84
RF baseline	0.88	0.82	0.84	0.83
RF hypertuned	0.36	0.23	0.17	0.09
DT baseline	0.86	0.78	0.80	0.79
DT hypertuned	0.88	0.81	0.83	0.82
CNB baseline	0.88	0.83	0.82	0.83
CNB hypertuned	0.88	0.83	0.82	0.83
GNB baseline	0.36	0.35	0.40	0.33
GNB hypertuned	0.68	0.68	0.77	0.65
MNB baseline	0.80	0.87	0.60	0.63
MNB hypertuned	0.83	0.77	0.76	0.76
KNN baseline	0.62	0.57	0.51	0.53
KNN hypertuned	0.62	0.56	0.56	0.56

Table 3.3: The average BoW classification results on the CARER dataset

exhibited no notable changes post-tuning, maintaining a lower performance level compared to other models.

In [112], the authors achieved their best accuracy with the BoW features using an LR model. In contrast, our work found that the SGD baseline model delivered the highest accuracy. Overall, the effect of hyperparameter tuning on model performance was mixed. While some models, like RF, experienced a drop in performance, possibly due to overfitting, others such as SVM, GNB, MNB, and DT - showed improvements across several metrics. On the other hand, models like SGD, LR, CNB, and KNN saw little to no change, indicating that tuning had minimal impact on their performance. This variability highlights the need for careful, model-specific hyperparameter optimisation.

3.3.3 Term Frequency-Inverse Document Frequency Feature Set

The use of TF-IDF feature representations offers a unique perspective on the efficacy of classification models, distinguishing itself from the purely syntactic approach of BoW. The results of the hyperparameter tuning for models using TF-IDF features demonstrate the optimal configurations for each model as determined by the hyperparameter search.

The hyperparameter search results for models using TF-IDF feature representations revealed that the SGD model with a learning rate of 0.0001, utilising a hinge loss function and an L1 penalty, would yield better performance. The SVM model was optimally configured with a regularisation parameter (C) of 10, a gamma value of 0.1, and a sigmoid kernel. For the LR model, the best performance was achieved with a C parameter of 10.0, paired with an L2 penalty and a liblinear solver. The RF model's optimal configuration included the use of the Gini impurity criterion for assessing the quality of splits, a maximum tree depth of 8, and 200 trees. The DT model was found to perform best with a maximum of 99 leaf nodes and a minimum of 2 samples required to split an internal node.

The CNB model was set with a smoothing parameter of 1.0, no class prior, fit prior set to true, and normalsation set to false. For the GNB model, the variance smoothing parameter was approximately set to 0.081. The MNB model was configured with a smoothing parameter of 0.5, no class prior, and fit prior set to false. Finally, the KNN model's neighbor parameter was set to 10. Table 3.4 highlights the performance of the classification algorithms using TF-IDF feature representations both before and after hyperparameter tuning.

Algorithm	**Accuracy**	**Precision**	**Recall**	**F1 Score**
SGD baseline	0.90	0.86	0.83	0.84
SGD hypertuned	0.90	0.87	0.84	0.85
SVM baseline	0.87	0.86	0.77	0.81
SVM hypertuned	0.89	0.85	0.82	0.84
LR baseline	0.88	0.87	0.78	0.81
LR hypertuned	0.89	0.86	0.84	0.85
RF baseline	0.89	0.84	0.83	0.83
RF hypertuned	0.36	0.21	0.17	0.10
DT baseline	0.86	0.80	0.81	0.80
DT hypertuned	0.61	0.74	0.63	0.61
CNB baseline	0.88	0.86	0.81	0.83
CNB hypertuned	0.88	0.86	0.81	0.83
GNB baseline	0.35	0.33	0.38	0.32
GNB hypertuned	0.60	0.64	0.69	0.59
MNB baseline	0.70	0.70	0.44	0.45
MNB hypertuned	0.83	0.82	0.70	0.74
KNN baseline	0.78	0.75	0.69	0.71
KNN hypertuned	0.82	0.81	0.71	0.74

Table 3.4: The average TF-IDF classificatoin results on the CARER dataset

The SGD algorithm demonstrated high accuracy and notable improvements in precision, recall, and F1 score post-tuning. This suggests that the fine-tuned parameters were well-suited to the algorithm's processing of TF-IDF features, enhancing its overall classification capability. Similarly, the SVM model also showed improvements in performance metrics after hyperparameter tuning. This indicates that the adjustments made to its configuration effectively enhanced its ability to classify the textual data more accurately.

The LR model exhibited increases in recall and F1 score following hyperparameter tuning. Conversely, the RF model experienced a significant decline in performance across all metrics after tuning. This drop could be attributed to an overfitting issue or an inadequate choice of hyperparameters. The DT model also showed a decrease in performance post-tuning, although its precision remained above 70%. The KNN model demonstrated a moderate improvement across all performance metrics after hyperparameter tuning.

For the Naive Bayes classifiers, while the CNB model showed minimal change post-tuning, the GNB algorithm exhibited considerable improvements across all metrics. Similarly, the MNB model also showed marked improvements, highlighting the impact of the smoothing parameter and the fit prior settings on its performance.

Overall, hyperparameter tuning generally improved model performance across various emotions, particularly in terms of precision. SGD, LR, and RF consistently ranked among the top performers across multiple emotions. However, GNB's performance varied significantly, demonstrating strong metrics in some cases but weaker in others. Compared to the results reported by Sundaram et al. [113], who yielded0.85 and 0.83 accuracy respectively, our hyperparameter-tuned SVM and baseline RF models outperformed their baseline model.

3.3.4 Distilled Bidirectional Encoder Representations from Transformers

Building upon the advancements of syntactical analysis, the fine-tuned DistilBERT model unlocks enhanced understanding of complex semantic relationships by generating contextualised representations of words.

Training configurations for the DistilBERT model included a batch size of 64, a weight decay of 0.01, and a learning rate of 2e-5. Initial training was completed using 10 epochs; however, the validation loss and training loss plotted in Figure 3.2 show the model is stable until the second epoch. The training loss continues to decrease, indicating the model is learning the training data well. However, after the second epoch, the validation loss starts to increase, indicating the model is overfitting the data. This divergence suggests that the model is starting to capture noise and details specific to the training set rather than learning generalisable features.

To address the observed overfitting, early stopping was implemented to halt the training

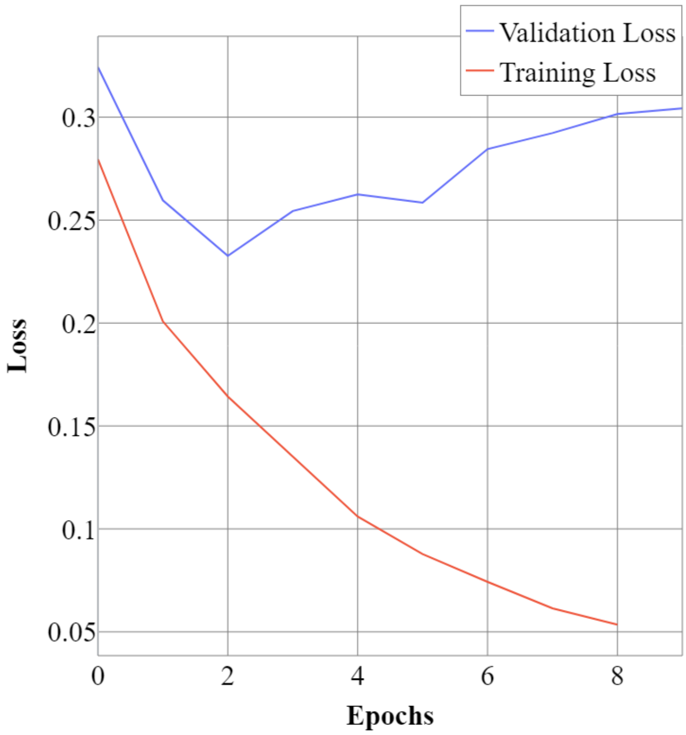


Figure 3.2: DistilBERT training and validation loss

process at the second epoch. As the scope of this work is to focus on model efficiency, the training results were acceptable, as the model can generalise sufficiently on unseen data based on the validation loss. Table 3.5 shows the classification results of the model after the training phase, as well as a confusion matrix in Figure 3.3.

Class	**Accuracy**	**Precision**	**Recall**	**F1 Score**
Anger	0.89	0.91	0.89	0.90
Fear	0.91	0.85	0.91	0.88
Joy	0.90	0.94	0.90	0.92
Love	0.81	0.72	0.81	0.76
Sadness	0.93	0.92	0.93	0.93
Surprise	0.62	0.67	0.62	0.65

Table 3.5: DistilBERT classification results on the CARER dataset

The model performs well when detecting the anger class, showing high precision and recall, indicating the model is accurate and successful in minimising false positives and

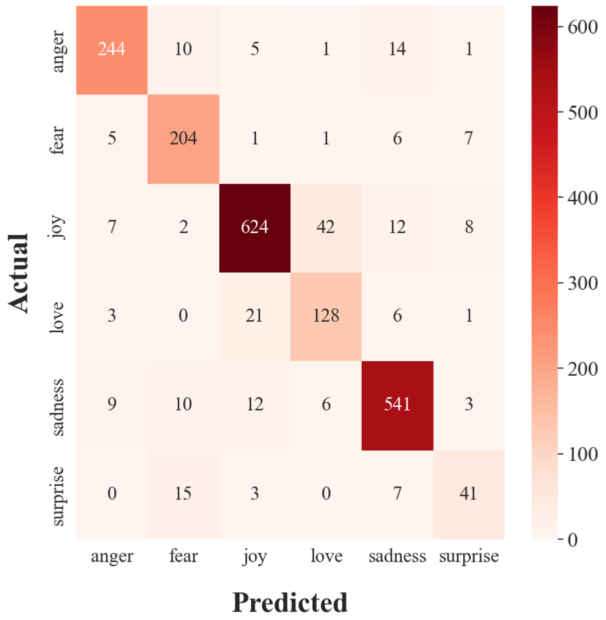


Figure 3.3: Confusion matrix for DistilBERT emotion classifier

negatives. For fear, the model shows a high recall but slightly lower precision, which implies the model can improve in reducing false positives. The model excels at identifying the joy class, demonstrating high precision and a respectable F1 score, which portrays the model's balance between precision and recall.

The model struggles comparatively when identifying the love class. While the recall is reasonable, a lower F1 score and precision suggest further refinement is needed for better performance in detecting the love emotion. Sadness is another strong class for the model, with both precision and recall scoring high, resulting in an excellent F1 score which reflects the model's consistency and reliability for this class. The surprise class poses the most significant challenge for the model, with the lowest scores across all metrics. These metrics show the model misses many instances of this class and indicate a need for improvement.

3.3.5 Model Latency

Model inference took place locally on a Raspberry Pi 4B using the Python Scikit-Learn package [111] for the ML classifiers and the Hugging Face platform transformers and pipeline packages [114] for the DistilBERT model. To measure the model latency, one test item from the test set was inferred and the completion time was recorded. Table 3.6 shows the latency time for each model in milliseconds.

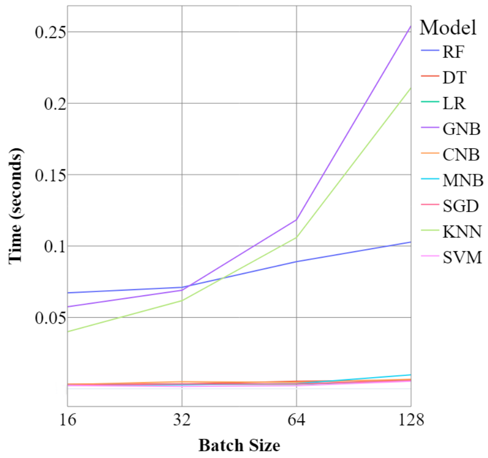
Model	TF-IDF Latency	BoW Latency
SGD	2.16 ms	3.53 ms
SVM	1.50 ms	1.34 ms
LR	2.85 ms	16.12 ms
RF	37.18 ms	17.28 ms
DT	3.12 ms	1.03 ms
CNB	2.09 ms	8.99 ms
GNB	17.85 ms	10.55 ms
MNB	2.18 ms	5.12 ms
KNN	17.69 ms	4117.71 ms

Table 3.6: Model inference latency in milliseconds (ms) using the CARER dataset

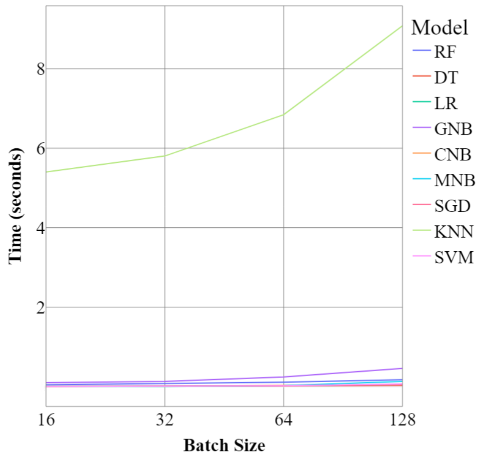
The SGD model performed efficiently for both feature representations, with just over a 1-millisecond difference. SVM displayed the fastest inference time for models using TF-IDF features and the second fastest for BoW features, indicating its efficiency in handling both feature representations. The LR model handled the TF-IDF features well but took significantly longer with the BoW features. The RF model demonstrated disparity in inference times between the two feature representations, performing better using BoW features. In contrast, the DT model demonstrated efficiency and recorded the fastest inference time of any model using the BoW feature representations.

CNB and MNB had comparable performance using TF-IDF features, each experiencing a latency of 2 milliseconds; however, with BoW features, the CNB model was over 1 millisecond quicker. The GNB model's inference time was longer when using TF-IDF features, performing moderately on both feature sets. The KNN model demonstrated a significantly larger inference time when using BoW features, which contrasted sharply with its performance when using TF-IDF features. Finally, the DistilBERT model, using contextual embeddings, showed a substantially higher latency of 251.9ms. This reflects the complexity of the model in comparison to traditional ML models, where less time was needed to make a prediction.

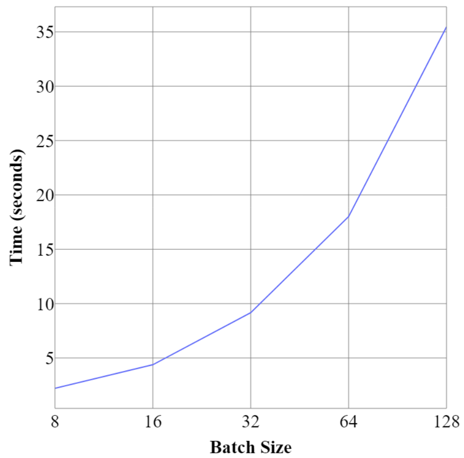
To measure latency in batches, batch sizes of 8, 16, 32, 64, and 128 were passed into the model, the results of which can be viewed in Figure 3.4. All measurements were recorded using a Raspberry Pi model 4B.



(a) TF-IDF batch latency



(b) BoW batch latency



(c) DistilBERT batch latency

Figure 3.4: Comparison of model batch latency

The analysis reveals that when using BoW feature representations, as shown in Figure 3.4b, DT and SVM consistently demonstrate the lowest latency, processing smaller batch sizes within approximately 5 milliseconds. LR and SGD models also exhibit relatively low latency for smaller batch sizes. Naturally, as batch sizes increase, the processing times for all models increase, with KNN displaying the most significant increase in time (9 seconds) for a batch size of 128. GNB exhibits higher processing times compared to other models, while CNB and MNB show moderate latency, making them less efficient for large-scale processing compared to other models. KNN demonstrates the highest latency among all models, rising up to 9.1 seconds for the largest batch size.

Notable variations in latency are visible when using TF-IDF features. Similar to BoW features, the SVM model consistently showed the lowest latency for all batch sizes, with processing times as low as 1.66 milliseconds for a batch size of 32. SGD and LR models also exhibited low latency. DT and CNB models demonstrated moderate latency, with processing times between 3 and 7 milliseconds across all batch sizes, with MNB performing similarly but exhibiting a higher processing time of 9.78 milliseconds for a batch size of 128. GNB and RF models displayed significantly higher processing times, increasing to 254.27 milliseconds and 102.8 milliseconds respectively for a batch size of 128. The KNN model showed the most significant increase in processing time with larger batch sizes, culminating in 2010.88 milliseconds for a batch size of 128.

The trend of increasing latency time with batch size continued with the DistilBERT model, as displayed in Figure 3.4c, with latency time rising almost linearly as the batch size doubles. For a lower batch size of 8, the model achieved 2.21 seconds, which almost doubled with a batch size of 16 (4.38 seconds). A more significant increase is observed with a batch size of 64, yielding a latency time of 18.01 seconds. The model recorded the highest latency time when testing a batch size of 128, which resulted in a processing time of 35.46 seconds.

3.3.6 Model Generalisation

To assess the performance of each ML classifier on diverse datasets containing different types of text and a broader range of emotions, two additional datasets were evaluated using the hyperparameters obtained from the CARER dataset grid search [100]. One of these datasets is the ISEAR dataset [115], a widely-used but older dataset based on student reports of situations that triggered emotions such as joy, fear, anger, sadness, disgust, shame, and guilt. Unlike the CARER dataset, ISEAR features more formal text with no slang and is a balanced dataset containing 7,666 entries.

The other dataset used was the DailyDialog dataset [116], which consists of multi-turn dialogues reflecting everyday communication on various topics. This is the most recent dataset used in this study and contains emotions like anger, disgust, fear, happiness, sadness, and surprise. The text in DailyDialog is more informal, often including slang and is inbalanced, consisting of 13,188 entries. Table 3.7 presents the classifiers' performance using each dataset and feature representation.

Model	Dataset	Accuracy	Precision	Recall	F1 Score
SGD BoW	DailyDialog	0.89/0.88	0.54/0.50	0.32/0.24	0.38/0.28
	ISEAR	0.55/0.56	0.55/0.55	0.55/0.56	0.55/0.56
SGD TF-IDF	DailyDialog	0.88/0.88	0.39/0.39	0.25/0.29	0.28/0.33
	ISEAR	0.53/0.52	0.54/0.52	0.54/0.52	0.54/0.52
SVM BoW	DailyDialog	0.88/0.88	0.57/0.50	0.34/0.24	0.40/0.28
	ISEAR	0.53/0.56	0.53/0.56	0.53/0.56	0.53/0.56
SVM TF-IDF	DailyDialog	0.88/0.88	0.44/0.44	0.29/0.29	0.33/0.33
	ISEAR	0.52/0.50	0.52/0.50	0.52/0.50	0.52/0.50
LR BoW	DailyDialog	0.88/0.88	0.56/0.57	0.29/0.25	0.36/0.30
	ISEAR	0.55/0.56	0.56/0.56	0.56/0.56	0.56/0.56
LR TF-IDF	DailyDialog	0.88/0.88	0.44/0.44	0.28/0.28	0.32/0.32
	ISEAR	0.52/0.52	0.52/0.52	0.52/0.52	0.52/0.52
RF BoW	DailyDialog	0.88/0.87	0.46/0.12	0.28/0.14	0.33/0.13
	ISEAR	0.54/0.49	0.54/0.56	0.54/0.49	0.53/0.49
RF TF-IDF	DailyDialog	0.87/0.87	0.12/0.12	0.14/0.14	0.13/0.13
	ISEAR	0.48/0.48	0.56/0.56	0.49/0.49	0.49/0.49
DT BoW	DailyDialog	0.86/0.87	0.44/0.24	0.39/0.20	0.41/0.21
	ISEAR	0.45/0.44	0.45/0.49	0.45/0.45	0.45/0.46
DT TF-IDF	DailyDialog	0.87/0.87	0.24/0.24	0.20/0.20	0.21/0.21
	ISEAR	0.45/0.45	0.53/0.53	0.45/0.45	0.46/0.46
CNB BoW	DailyDialog	0.80/0.80	0.30/0.30	0.42/0.42	0.33/0.33
	ISEAR	0.57/0.57	0.56/0.56	0.57/0.57	0.56/0.56
CNB TF-IDF	DailyDialog	0.80/0.80	0.30/0.30	0.42/0.42	0.33/0.33
	ISEAR	0.52/0.52	0.52/0.52	0.53/0.53	0.52/0.52
GNB BoW	DailyDialog	0.07/0.07	0.21/0.21	0.53/0.53	0.10/0.10
	ISEAR	0.33/0.39	0.33/0.42	0.33/0.40	0.32/0.38
GNB TF-IDF	DailyDialog	0.07/0.07	0.21/0.21	0.53/0.53	0.10/0.10
	ISEAR	0.32/0.39	0.33/0.42	0.33/0.40	0.32/0.38
MNB BoW	DailyDialog	0.87/0.75	0.20/0.27	0.16/0.42	0.17/0.31

Model	Dataset	Accuracy	Precision	Recall	F1 Score
	ISEAR	0.56/0.53	0.57/0.53	0.57/0.53	0.57/0.53
MNB TF-IDF	DailyDialog	0.87/0.75	0.20/0.27	0.16/0.42	0.17/0.31
	ISEAR	0.54/0.53	0.54/0.53	0.54/0.53	0.54/0.53
KNN BoW	DailyDialog	0.88/0.87	0.29/0.21	0.18/0.15	0.20/0.14
	ISEAR	0.33/0.32	0.36/0.33	0.31/0.31	0.31/0.29
KNN TF-IDF	DailyDialog	0.88/0.87	0.29/0.21	0.18/0.15	0.20/0.14
	ISEAR	0.30/0.30	0.33/0.33	0.31/0.31	0.29/0.29

Table 3.7: Classification results using the ISEAR and DailyDialog datasets (Baseline/Hypertuned)

The SGD model maintained consistent performance across both datasets and feature types, showing a slight improvement in precision and recall on the DailyDialog dataset with TF-IDF features. The LR model achieved minor precision and F1 improvements on ISEAR, particularly with the BoW representation. In contrast, the RF model showed decreased performance across both datasets and feature representations, similar to the results on the CARER dataset. For the DT model, tuning yielded only limited gains.

The CNB model demonstrated stable metrics across feature types and datasets, while the GNB model achieved a modest improvement on the ISEAR dataset. Although MNB exhibited a significant drop in accuracy on DailyDialog, it saw gains in recall and F1 score. Lastly, the KNN model showed minimal improvements across all metrics.

In conclusion, applying these hyperparameters yielded mixed results across models and datasets. Performance on the ISEAR dataset remained generally low despite hyperparameter adjustments, indicating that the selected models and features may not be well-suited for this type of text. In contrast, results on the DailyDialog dataset were higher and comparable to those on the CARER dataset. Although the datasets differ in text type, DailyDialog shares similarities with CARER, including an imbalanced structure, which suggests these techniques are more suitable for imbalanced data with informal language. However, the hyperparameters may still require further optimisation for these datasets.

3.3.7 Discussion

The comprehensive analysis of various ML models using BoW, TF-IDF feature representations, and DistilBERT has provided valuable insights into the efficiency of these models in classifying emotions on an edge device. The recorded training times demonstrate that models generally train faster with TF-IDF feature representations compared to BoW, suggesting that TF-IDF is more computationally efficient. This could be due to the dimensionality of vectors used in the BoW features (16182) when using the dataset by [100].

The models that exhibited the fastest training times included KNN, CNB, and MNB models using TF-IDF features, whereas RF and DT models using BoW features were the slowest. SGD, SVM, and LR models showed moderate training times with TF-IDF features but experienced significant increases with BoW. The fine-tuning of the DistilBERT model could not be performed on the Raspberry Pi 4, highlighting the need for an edge board with greater compute power to facilitate training on the edge.

When evaluating performance, SVM and SGD models with BoW features, and SGD and LR models with TF-IDF features, achieved the highest performance metrics. The SGD model is particularly effective with sparse data and updates model parameters incrementally, showing robustness with high-dimensional data as evident with TF-IDF and BoW features, which explains its strong performance across both feature sets. Poorly performing models included GNB, MNB, and KNN, although KNN demonstrated better performance with TF-IDF features. GNB is known for handling sparse data poorly, suggesting it may not be suitable in this scenario. MNB can also be sensitive to class imbalance, which is apparent in this dataset, reflecting in its results. The DistilBERT model consistently performed well across all metrics, underscoring its effectiveness for this task.

When it came to model inference, DT, SVM, and SGD models recorded some of the fastest times, whereas RF, DistilBERT, and KNN showed slower performance. KNN exhibits high computational complexity during the prediction phase due to its calculation of distances between training instances and testing data, resulting in prolonged processing times. RF's ensemble structure, which includes multiple decision trees, necessitates multiple rounds of data splitting during prediction, leading to slower latency times. The DistilBERT model, based on transformer architecture effective for understanding contextual relationships, requires substantial computational resources, as evidenced by the latency results. TF-IDF features proved more efficient than BoW across all models in terms of training time and latency.

Based on the analysis of performance and latency, it is evident that the SGD model with TF-IDF features emerges as the most efficient model. SVM with TF-IDF features closely follows, with slightly lower performance than SGD but marginally quicker inference times. The combination of efficiency and effectiveness makes the SGD model particularly valuable for practical applications where both accuracy and computational resources are crucial considerations.

The evaluation demonstrates that dataset imbalance significantly impacts model performance, particularly in underrepresented classes such as surprise and love, where limited training samples may have constrained the models' learning effectiveness. Interestingly, most models performed better with imbalanced datasets, highlighting the importance of selecting feature representations and models that accommodate specific dataset characteristics, especially in contexts involving informal language and slang. Furthermore, incorporating data collected by this system in a longitudinal study could offer valuable insights into model generalisation and robustness, particularly concerning feature dimensionality and varied dataset properties.

3.4 Conclusion

This work has conducted a thorough comparison of the performance of nine distinct ML classifiers, specifically evaluating their effectiveness with BoW, TF-IDF feature representations, and DistilBERT. The findings from this study highlight the crucial role of feature representation in influencing model performance and emphasise the importance of careful model and feature selection in the context of emotion classification in an edge architecture. Each combination of model and feature should be meticulously assessed based on its specific strengths to achieve optimal performance.

In this comparative analysis, BoW, TF-IDF, and DistilBERT all showed strong performance. However, the SGD model combined with TF-IDF features emerged as the most efficient. Overall, models utilising TF-IDF representations outperformed those using BoW, displaying more consistent and balanced performance across various metrics. This trend was also observed during model inference, where the high dimensionality of BoW features notably influenced performance. These findings underscore the intricate relationship between feature representations and classifier effectiveness, offering valuable insights for future research on emotion recognition in textual data, particularly in environments with limited computational resources where efficiency is critical. They demonstrate that traditional ML approaches to text classification remain viable options over more complex solutions.

Building on the findings of this chapter, subsequent work incorporates these insights into a broader multimodal framework. Chapter 4 shifts focus to the visual domain, evaluating the feasibility of thermal facial expression recognition using low-cost hardware and deep learning architectures. This chapter complements the textual analysis by exploring another resource-conscious modality suited for privacy-preserving, real-world deployment. Chapter 5 then integrates these modalities - text, thermal, and speech - using both feature-level and decision-level fusion strategies. The use of TF-IDF representations and efficient classifiers are further validated in real-time multimodal systems tested on constrained hardware. These developments demonstrate the practical viability of traditional machine learning approaches for textual emotion classification as part of a scalable, multimodal emotion recognition system designed for deployment in resource-limited environments.

Chapter 4

Thermal Facial Expression Emotion Recognition

4.1 Introduction

FER is a crucial area of research within the fields of affective computing and computer vision. It involves the identification of human emotions based on facial cues or poses, which are essential for non-verbal communication. Starting in the 1970s, Paul Ekman and Wallace V. Friesen categorised expressions into six basic emotions: happiness, sadness, anger, fear, disgust, and surprise [117]. Nowadays, FER has become much more sophisticated by incorporating specialised hardware and a variety of techniques, including local feature analysis [118], holistic analysis [119], and DL techniques such as CNNs [120].

Although the CNN architecture has predominantly dominated computer vision tasks such as image classification and object recognition [121], recent advancements in ML, particularly the introduction of ViTs [122], have enabled FER systems to become more sophisticated and robust. Early research shows that ViTs often demonstrate superior performance over CNNs due to their lack of locality inductive bias, which allows them to capture global dependencies more effectively and leverage self-attention mechanisms to provide a more holistic representation of input data [123].

The intersection of emotion recognition technology and affordable hardware solutions has sparked significant interest within the research community. Traditionally, FER has required expensive, specialised equipment often only available in a commercial environment.

However, the advent of affordable hardware capable of delivering comparable performance to costly alternatives is reshaping the landscape of ML applications [124]. Moreover, the rapid development of IoT technology has inspired the utilisation of low-resolution cameras as a viable, cost-effective solution for different types of computer vision tasks [125]. Low-resolution imagery, once considered a hindrance, can now be harnessed effectively thanks to advancements in ML architectures. However, images in the visible spectrum pose privacy and security risks and often do not meet the ethical requirements for real-world applications. Therefore, in this work, we utilise a thermal camera, which is considered non-invasive.

Thermal imaging offers several key advantages over visible spectrum cameras in FER tasks. First, thermal cameras are not affected by lighting variations, ambient light conditions, or shadows [126]. Second, thermal imaging captures more nuanced details related to physiological signals, such as heat distribution and blood flow, which correlate with emotional states and provide additional visible cues not present in RGB cameras [127]. Third, thermal data protects identity more effectively than RGB images due to its inability to capture fine-grained texture and colour information, making it a more privacy-conscious choice for real-world applications [128]

This chapter explores the potential of these technological developments by introducing a new dataset for thermal emotion classification and performing a comprehensive performance comparison using ResNet50 [129], a state-of-the-art CNN model, against two ViT models; one by Google [130] and another modified architecture tailored for smaller dataset sizes [131]. We also form a hybrid dataset with the state-of-the-art KTFE [132] to provide insights into the generalisation potential of these technologies. To the best of our knowledge, no research papers include ViT models when conducting emotion classification using images in the thermal spectrum, and no thermal camera dataset exists where data is collected using an IoT-based thermal camera. Our key contributions are as follows:

- Introduction of the University of the West of Scotland Thermal Faces (UWSTF) novel FER dataset collected using low-cost thermal cameras
- Exploration of whether low-cost hardware can provide performance comparable to expensive hardware through state-of-the-art methods

- Investigation into the viability of using low-resolution cameras for emotion classification
- A side-by-side comparison of CNNs and ViT models on both our dataset and the hybrid dataset, addressing the question of bias produced by CNN models with vision transformers

The rest of this chapter is organised as follows: Section 4.2 describes the data collection methodology for the UWSTF dataset, data preprocessing steps, and model architecture, Section 4.3 details the results and describes the evaluation metrics and general discussion before the chapter is concluded in Section 4.4.

4.2 Materials and Methods

In this section, we explain the data collection methodology by providing information on how participants were recruited, how emotions were stimulated, what equipment was used, what the environment setup looked like, how the dataset was curated, what the preprocessing stages entailed, and information on the architecture of the CNN and ViT models.

4.2.1 Participant Recruitment

Fourteen male participants, aged between 21 and 42, were recruited. They were either post-doctorate researchers or PhD students at the University of the West of Scotland in the UK. The participants hailed from various countries, including Scotland, England, China, Romania, and Pakistan. They were recruited through internal communications within the research group, and the study was approved by the ethics board at the University of the West of Scotland under project #18323. None of the participants wore glasses.

Participants were assigned a specific time and date for data collection. Upon arrival, they were provided with the necessary information sheets and consent forms, which were signed before any exercises commenced.

In addition to recording age, gender, and nationality, participants were asked whether they had any known affective disorders or prior experience with tasks involving facial expression recognition. None reported any affective history or prior exposure to such tasks. While the participant pool consisted of individuals in a research setting, this background was not associated with specialised training in affective computing. These measures were taken to minimise potential selection bias and enhance the generalisability of the dataset within the scope of this exploratory study.

4.2.2 Stimuli

A series of video clips was selected by the authors to capture spontaneous facial expressions, which included 2 videos for anger, 2 for fear, 2 for happiness, 9 for neutral, 2 for sadness, and 2 for surprise. Videos have been found to be effective stimuli for eliciting a wide range of emotions [133, 134]. Short videos, such as YouTube Shorts and TikTok, are extremely popular and were used to keep participants engaged [135].

The selection process was informed by prior work in similar thermal datasets such as KTFE [83] and NVIE [84], and validated through participant self-reports immediately following each video clip. In KTFE, stimuli were selected by the authors based on cultural relevance, with participants self-reporting their emotional responses using standardised scales. In NVIE, emotional videos were used to elicit spontaneous expressions, and self-reported responses were used to confirm the appropriateness of each clip, with mismatches excluded.

Similarly, we chose videos based on cultural relevance and public viewer comments indicating strong emotional responses. After each clip, participants reported how the video made them feel, and only clips where the intended emotion was confirmed by participant feedback were retained in the dataset. While most emotions during spontaneous elicitation were successfully captured - particularly happiness and surprise - some emotions proved more difficult to evoke, as also reported in previous work [136].

For the posed facial expression data, emoticons representing each facial expression were displayed to prompt participants. Rather than imitating the emoticons directly, participants were asked to produce expressions that felt natural to them for each emotion category. Emoticons were selected for their simplicity and cultural familiarity, which have been shown to support intuitive emotional interpretation and expression [137].

4.2.3 Equipment

A TOPDON TC001 thermal camera was used in this study to collect thermal images. The TC001 operates in a spectral range of 8-14 micrometres and has a resolution of 256x192 pixels. The camera can record at a rate of up to 25 frames per second and has a temperature range of -20 to 150 degrees Celsius with a temperature accuracy of Âś2 degrees Celsius. The camera also features a Noise Equivalent Temperature Difference of 40 milli-Kelvin at 25 degrees Celsius and operates with a power consumption of 0.35W. The TC001 weighs 30 grams and measures 71x42x14 millimetres.

The camera was connected to an ASUS Vivobook laptop running Windows 11, which had the TCView software installed. A table was used to place the laptop, and a 50-inch Samsung TV was used to show video clips and images of the emoticons. A tripod was employed to hold the camera in position, and a traditional office chair was provided for participants to sit on.

4.2.4 Environment Setup

The data was collected inside a lab at the University of the West of Scotland. The room was an approximately 24 square metre L-shaped area, featuring four windows on one wall and five windows on another side. The room was not empty as there were various desks with computer monitors situated around the perimeter. A small 3x3 metre area in the room was used as the data collection area, the arrangement of which can be seen in Figure 4.1.

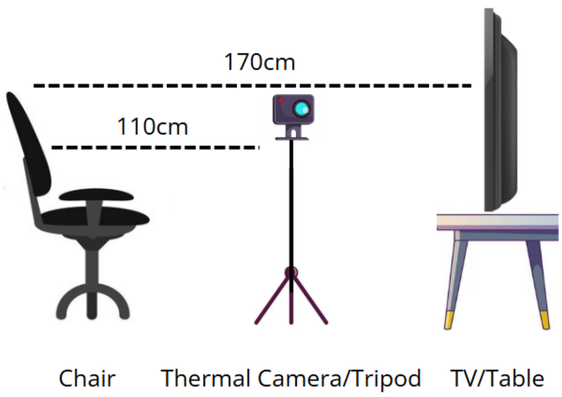


Figure 4.1: The approximate data collection set up

To ensure participant discretion and comfort, data collection was conducted in a dedicated and quiet section of the lab with minimal distractions. The 3x3 metre area used for recording was positioned away from windows and foot traffic. Blinds were drawn during data collection to reduce visual distractions and control natural light. Only essential personnel were present, and no observers were in the direct line of sight of participants.

Participants were seated alone during recordings and instructed to behave naturally, with the assurance that there were no 'correct' emotional responses. They were also informed that they could withdraw at any time without consequence. Room temperature and lighting conditions were kept consistent across all sessions to avoid external influences on emotional expression. The thermal camera setup was non-intrusive, and no visible facial markers or physical contact were used to preserve a neutral and comfortable environment.

4.2.5 Protocol

Spontaneous and posed facial expressions were collected as part of this study. Before the collection began, each participant was given information about the process and the option to opt out at any time. First, using the TC001 camera recording at 25 frames per second, spontaneous emotions were collected. This involved participants watching 19 short video clips, presented in a random order, on a TV with the camera placed between the TV and the participant. Starting with a sad short video followed by a neutral clip, this pattern continued by first playing a short video that elicited an emotion other than neutral, followed by a neutral video clip, an approach taken by [84] to help alleviate any negative feelings. After each short video clip, the participant reported how that clip made them feel from the six emotions: anger, fear, happiness, neutrality, sadness, surprise, or other. This is a similar approach to that taken by [132]. Once all short videos were watched and emotions were reported, the video recording of the participant was stopped.

After the spontaneous collection was complete, participants were asked if they were comfortable proceeding to the posed collection activity, to which all responded affirmatively. To record the posed emotions, the same setup was used as in the spontaneous collection, with the TC001 recording. However, rather than video clips playing, the participant was asked to mimic the emoticon shown on the TV. The participants were asked to mimic each posed expression in a random order, three times for three seconds each, reverting to a neutral pose between expressions. After the posed data collection was complete, each participant was debriefed to ensure that they were not affected by any of the materials used to induce the emotions. None of the participants reported feeling any adverse effects, concluding the data collection process.

4.2.6 Dataset Curation

Separate thermal video recordings were created for each emotion-eliciting clip, allowing individual emotional responses to be evaluated in isolation. After data collection was completed, the recordings were saved in MP4 format and prepared for frame extraction. This separation ensured that only frames corresponding to a specific emotional stimulus were considered during the annotation process. We adhered to the following process when collecting the data:

- Start video clip
- Start thermal recording
- Stop video clip
- Stop thermal recording
- Collect participant's self-reported emotion

To create the thermal facial expression dataset each spontaneous and posed video recording was exported into individual frames in JPG format using the OpenCV library in Python [138]. The selection process was performed manually by the authors, as shown in Algorithm 1. Every frame was reviewed and evaluated based on the participant's self-reported emotion following each video clip, as well as visible facial cues in the thermal image such as changes in cheek, eye, or mouth regions. Only frames that clearly reflected the intended emotional state were included in the dataset.

This manual annotation process follows similar practices in thermal FER literature. For example, [19] performed manual frame selection and verified annotations through expert agreement, while the KTFE dataset [83] relied on participant self-reports and controlled experimental protocols to guide the inclusion of expression samples. Each selected image

Algorithm 1 Thermal Frame Annotation Procedure

```
for each thermal video clip do
    Convert video to image frames
    if participant's reported emotion matches intended emotion of clip then
        for each extracted frame do
            if thermal facial cues are visibly aligned with reported emotion then
                Add frame to dataset with corresponding emotion label
            end if
        end for
    end if
end for
```

was then placed into a directory corresponding to one of six emotion classes and prepared for preprocessing.

As a result, a total of 61 images for anger, 130 images for happiness, 97 images for neutrality, 92 images for sadness, and 178 images for surprise. Figure 4.2 shows an example of each expression from the exported frames.

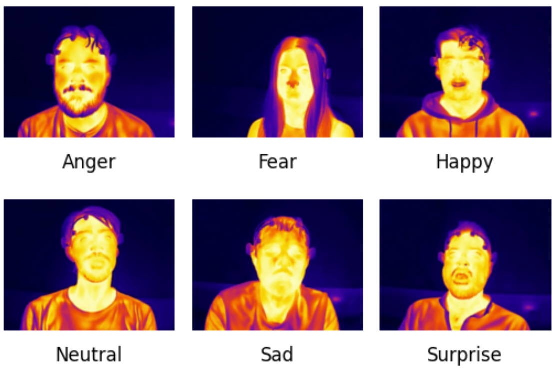


Figure 4.2: An example image of each emotion in the dataset

4.2.7 Preprocessing

Various steps were taken to prepare each image in the dataset for model training. First, a Python script was created in which the images were loaded from the directories previously created during the dataset curation step using the OpenCV library [138]. Next, a pre-trained Haar Cascades classifier was used on each image to detect whether a face was present. As expected, due to the manual diligence in the dataset curation step, a face was successfully found in each image.

Using the coordinates of the face's position identified by the Haar Cascades model, the face was cropped from each image, the remaining background was removed, the image was converted to greyscale, and finally normalised. The image was then resized to 48x48, which are common dimensions for many types of image classification tasks, as it is large enough for models to pick up nuances and small enough to be efficient for the training stage and inference. The channels of the image were then merged to ensure the image had three channels, as this format is required by certain models. Finally, the dataset was split into training, validation, and testing sets using a 60/20/20 ratio.

4.2.8 Convolutional Neural Network

CNNs are DL models primarily used for structured grid data, such as images, using a series of layers that include convolutional layers, pooling layers, and fully connected layers [139]. CNNs employ kernels in the convolutional layers that slide over the input data to extract local features, which are crucial for the model in recognising patterns in images. These fixed patterns can be a limitation of the CNN model, but they save training time as the model doesn't need to learn how to focus.

This process is followed by activation functions, typically Rectified Linear Units (ReLU), which introduce non-linearity into the model, enabling it to learn complex patterns. To reduce the spatial dimensions of feature maps, pooling layers are used, often employing max pooling or average pooling. These layers decrease the computational load and help prevent overfitting. This hierarchical structure enables CNNs to learn increasingly abstract features with each layer, from simple edges in the initial layers to more complex shapes and objects in the deeper layers [140].

CNNs have gained prominence in FER due to their ability to learn features from facial images without the need for manual feature extraction, which was a limitation of traditional ML methods. The architecture of CNNs allows them to effectively capture both low-level features, such as basic textures and edges, and high-level features, such as facial expressions. Recent advancements in DL have significantly improved the performance of FER models, making them operate effectively in challenging conditions such as variations in lighting, head pose, and angles [141].

To further enhance the performance of CNNs, some approaches leverage transfer learning, where pre-trained CNN models are fine-tuned on FER datasets, allowing for better generalisation and performance on new data [142]. Overall, CNNs combine their robust feature extraction capabilities with advanced training techniques to achieve high accuracy and reliability, representing a powerful tool for recognising human emotions from facial expressions. In this work, we fine-tune the ResNet50 pre-trained model [129] using the Keras DL API [143].

4.2.9 Vision Transformers

The introduction of the transformer model [144] has significantly influenced NLP and has recently advanced into vision and multimodal applications. In recent years, state-of-the-art image classification has been dominated by CNNs until the introduction of ViT models. Unlike CNNs, which rely on local receptive fields and hierarchical feature extraction, ViTs utilise a self-attention mechanism. This mechanism allows the model to capture global dependencies across the entire image by dividing the image into patches, which are then linearly embedded into a sequence of tokens for the transformer to process [122]. The self-attention mechanism enables the model to assess the importance of different patches, facilitating a better understanding of spatial relationships and contextual information within the image.

ViTs have shown promising performance in the field of FER, especially in challenging scenarios such as varying lighting conditions and occlusions. The authors in [145] present TFE (Transformer Architecture for Occlusion Aware FER), demonstrating that transformers can effectively manage occlusions by focusing on relevant facial features while ignoring irrelevant background noise. Additionally, although different types of ViT models exist, [146] employs the Mask Vision Transformer (MVT) model, which incorporates a mask generation network to filter out complex backgrounds and occlusions, a technique that further enhances the model's robustness, especially for FER in the wild.

Moreover, ViTs address limitations inherent in CNNs. Traditional CNNs rely on local features and often struggle with fixed-size input requirements [147]. The global receptive field of transformers allows them to learn more comprehensive representations of facial expressions by utilising positional embeddings, enabling the model to capture nuances within the data in unexpected ways. However, this comes at a cost: the training time increases, and larger amounts of training data are required, resulting in high computational resource demands. Research using ViT models for thermal FER is not as prevalent as work involving CNNs. In this study, we aim to address this gap by comparing the performance of a modified ViT model described in [131], herein referred to as the modified ViT, designed for training on smaller datasets utilising the Keras API, and fine-tuning on a pre-trained ViT model developed by researchers from Google [130] using the Hugging Face platform [13].

4.3 Results

In this section, we describe the feature extraction process and the chosen evaluation metrics to assess the models' performance. We compare the training and testing performance and provide insights into generalisation capabilities by incorporating test data from the KTFE dataset.

4.3.1 Feature Extraction

To evaluate the performance of the CNN architecture, the feature extraction process utilised the pre-trained ResNet model's convolutional layers as a foundational component. This approach processes input images by leveraging the model's pre-trained weights on the ImageNet dataset [148]. The outputs from the convolutional layers capture high-level abstract representations of the input images, which are then passed to subsequent layers for further processing. These layers focus on feature refinement and classification based on the extracted features. Some studies have shown that using regions of interest can enhance the performance of CNNs [149], whereas other work emphasises the importance of using the whole face to capture temperature changes across the facial surface [150].

The pre-trained ViT model developed by Google extracts features following the standard architecture of the model. First, it divides the input image into non-overlapping patches of 16x16 pixels. Patch embeddings are created by flattening each patch before linearly projecting them into a vector. Positional embeddings are added to the patch embeddings to incorporate spatial information. These embeddings serve as input tokens for the Transformer, where the core feature extraction occurs within the Transformer encoder, which comprises several layers. To learn intricate relationships between patches, each layer features multi-head self-attention, a feed-forward neural network (Multi-Layer-Perceptron) for non-linear feature extraction, and layer normalisation to stabilise the network, resulting in a sequence of rich token embeddings.

The modified ViT model also follows the standard architecture of the model in the feature extraction process; however, it implements further customisations. For instance, it tokenises patches using shifted patch tokenisation, whereas the Google model utilises a simpler patch embedding mechanism based on fixed-size non-overlapping patches. Furthermore, the Google model relies on standard multi-head attention, whereas the modified model introduces a trainable temperature parameter to scale the query vectors, providing more flexibility compared to the standard multi-head attention mechanism. Despite these variations, both ViT models utilise the transformative power of attention mechanisms to capture complex relationships for effective feature extraction.

4.3.2 Evaluation Metrics

To evaluate the effectiveness of the models, the following performance metrics were included: accuracy, precision, recall, and F1 score. Confusion matrices were also utilised to provide insights into accuracy and error patterns across the different emotion categories.

- Accuracy - a straightforward metric that offers an overall sense of model performance by measuring the proportion of correctly classified instances over the total number of instances.
- Precision - a metric that reflects the model's ability to identify positive samples correctly by measuring the ratio of true positive instances to the sum of true positive and false positive instances.
- Recall - a metric that evaluates the model's capability to capture all relevant instances by measuring the ratio of true positive instances to the sum of true positive and false negative instances.
- F1 Score - a metric that balances the importance of precision and recall by measuring the harmonic mean of precision and recall.

4.3.3 Comparison of Model Training

To train each model, the data was split into 60/20/20 for training/validation/testing. Each model underwent 200 epochs of training; however, early stopping callbacks were implemented to prevent overfitting. These callbacks were based on the evaluation of accuracy and loss from the training and validation data and included a function to reduce the training learning rate when a plateau was detected.

Starting low but gradually increasing over time, the training process for the ResNet model lasted for 125 epochs before stopping early. Figure 4.3 illustrates that the validation accuracy initially surpassed the training accuracy, suggesting effective generalisation to the validation data.

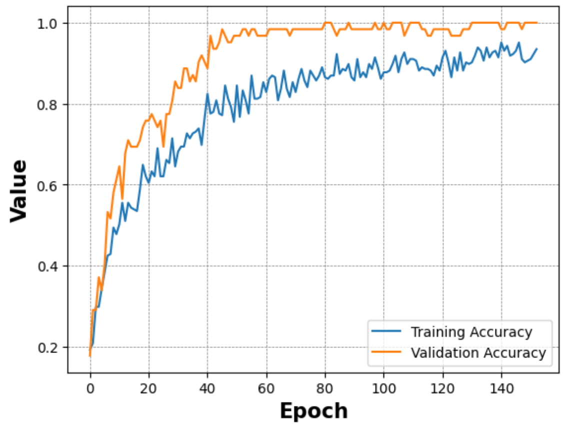


Figure 4.3: ResNet training and validation accuracy

The ViT model by Google exhibited a rapid learning process, as shown in Figure 4.4, achieving near-perfect accuracy within just a few epochs. In contrast to the ResNet model, the training data accuracy surpassed the validation data accuracy early on, with both following similar trajectories. This indicates that the model learned quickly and overfitted to the training dataset, suggesting that the early stopping patience was insufficient to prevent overfitting. However, the validation accuracy was still respectable, highlighting the model's generalisation capabilities.

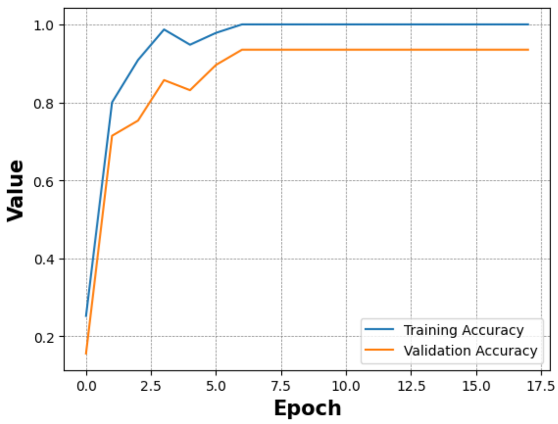


Figure 4.4: ViT by Google training and validation accuracy

Figure 4.5 shows the modified ViT model, which demonstrated a swift learning process, terminating training early at 46 epochs. Despite minor fluctuations, both training and validation accuracies stabilised early and maintained high accuracy, indicating efficient generalisation and adaptation.

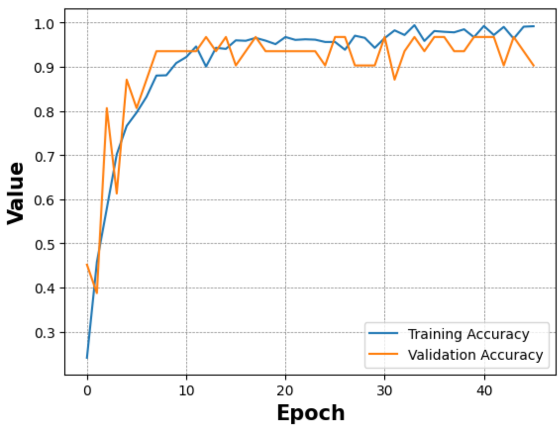


Figure 4.5: Modified ViT model training and validation accuracy

Overall, each model displayed different behaviours in the training process, highlighting their unique characteristics and learning speeds. Notably, the ViT models managed to learn in far fewer epochs than the ResNet model.

4.3.4 Comparison of Model Performance

After the training process, the models' performance was evaluated using the test data from the training dataset split, with results reported using the metrics described in Section 4.3.2.

Macro averages were used to provide insights across class distributions. The results are summarised in Table 4.1 while Table 4.2 shows the results of each model at a more granular level for each class.

Model	**Accuracy**	**Precision**	**Recall**	**F1 Score**
ResNet	95	92	96	94
Google ViT	96	96	95	95
Modified ViT	96	97	96	96

Table 4.1: Each Model's Performance for Accuracy, Precision, Recall, and F1-Score

The ResNet model demonstrated high accuracy at 95. Precision reached 92, indicating a respectable rate of true positive predictions. Recall was slightly higher at 96, suggesting that the model effectively identified the majority of positive instances, resulting in an F1 score of 94.

The model achieved perfect scores when predicting Fear and Surprise; however, it showed some limitations in classifying Neutral and Anger emotions. While precision and F1 Score for Anger were high, the recall was lower at 86, whereas when classifying the Neutral emotion, precision was relatively low but F1 Score and recall were high. Similarly, although precision was high for the Sad emotion, it was the lowest scoring metric with high F1 score and perfect recall.

Model	Metric	An	Fe	Ha	Ne	Sa	Su
ResNet	Precision	92	100	100	75	88	100
	Recall	86	100	86	100	100	100
	F1	92	100	93	86	93	100
Google ViT	Precision	100	89	100	89	100	96
	Recall	100	89	100	89	93	100
	F1	100	89	100	89	96	98
Modified ViT	Precision	100	100	085	100	100	100
	Recall	100	100	100	82	93	100
	F1	92	100	92	90	97	100

Table 4.2: The performance of each model for each class

The ViT model by Google achieved slightly higher accuracy than the ResNet model, reaching 96. The model also achieved considerably more precision at 96. Finally, with a recall of 95, the model maintained effective detection capabilities, albeit slightly less sensitive than its precision, culminating in an F1 score of 95. Unlike ResNet, this model managed to identify the Anger and Happy emotions with perfect precision, recall and F1 scores. Performance deteriorated when identifying Fear and Neutral classes, although metrics still remained high. For the Sad emotion, the model presented excellent precision of 100 and respectable recall and F1 score, and when predicting Surprise, the model performed exceptionally well, closely approaching perfect scores.

The modified ViT model excelled and surpassed ResNet and Google ViT in almost all metrics. Although the same accuracy score was achieved as the model by Google, the modified ViT achieved 97 for precision, demonstrating a higher proportion of correct positive predictions. Its recall matched the accuracy score at 96, resulting in an F1 score of 96, marking the modified ViT as the top-performing model in our comparison. Similar to ResNet, this model achieved perfect scores for the Surprise and Fear classes, while detecting Anger was also near-perfect besides the F1 score of 92. The model experienced some challenges with the Happy emotion where model precision and F1 score were lower despite a perfect recall score. This model achieved the highest precision for Neutral, yet recall slightly lagged, resulting in an F1 score of 90. This was similar to the Sad emotion;

however, recall was higher, which resulted in an F1 score of 97.

In summary, while all models exhibited high performance across metrics, ResNet showed strong general performance, particularly for Fear and Surprise classes, while the ViT by Google excelled with Anger and Happy recognition. The modified ViT revealed areas for improvement but demonstrated impressive precision for Surprise and Sad emotions. The modified ViT outperformed the others, particularly excelling in overall precision and F1 score, underscoring its efficacy in thermal emotion classification.

4.3.5 Model Generalisation

In our study, we evaluated the generalisation capabilities of thermal emotion classification using learning curves - a method that provides insight into a model's performance as the training dataset size gradually increases [151]. We tested our models on different portions of the dataset, ranging from 50% to 100%. Our evaluation focused on two scenarios: using only the KTFE dataset and using a mixed dataset comprising KTFE test samples alongside a 20% test split from our own collected dataset.

For the KTFE dataset, 90 samples were selected as a fixed test set across all generalisation experiments. The remaining 359 images were used for training, with subsets ranging from 50% to 100% of the available training data. The KTFE dataset contained 449 images. In the mixed dataset configuration, an additional 106 test samples from our own dataset were included alongside the KTFE test items and these test sets remained unchanged across all training set size comparisons. To ensure reproducibility and consistent evaluation, a fixed random seed (42) was used during preprocessing.

Although some models showed signs of overfitting during training, early stopping was implemented based on validation accuracy and loss, and the best-performing model checkpoint on the validation set was restored for evaluation.

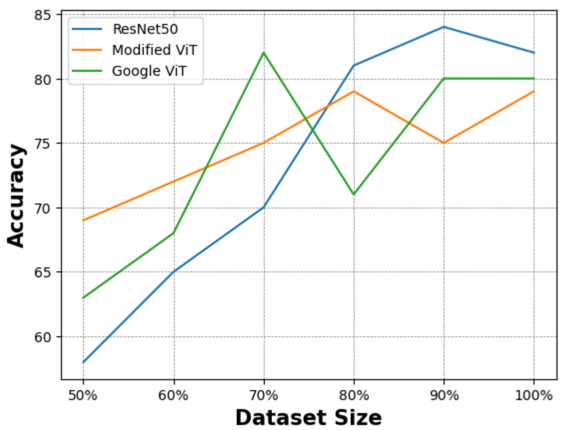


Figure 4.6: Model accuracy on different sizes of the KTFE dataset

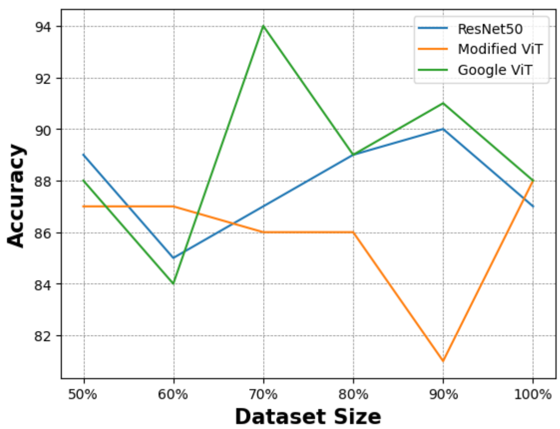


Figure 4.7: Model accuracy on different sizes of the mixed dataset

The ResNet model achieved an initial accuracy of 58% when half of the KTFE dataset was utilised, which increased steadily as more of the training data was incorporated, eventually peaking at 84% with 90% of the dataset - the highest accuracy recorded for the KTFE test data - before dropping slightly to 82% when the full dataset was utilised. In contrast, testing accuracy on the hybrid dataset remained consistently around the high 80s and 90% mark, starting at 89% when using 50% of the data, suggesting strong model generalisation capabilities.

The Google ViT model's results fluctuated when testing on the KTFE dataset, starting at 63% accuracy before sharply increasing to 82% when using 70% of the dataset as part of the training data. This was subsequently followed by a slight dip in performance, but it remained stable at around 80% when larger portions of the dataset were used. In the mixed testing data, the model started with an impressive 88% accuracy using only 50% of the KTFE dataset in the training data before reaching a high of 94% with 70% of the training data used - the highest accuracy recorded - followed by a slight decrease in performance but remaining above 88%.

When using the modified ViT model, accuracy on the KTFE test set started at 69% with 50% of the KTFE dataset used in the training data before peaking at 79% when using 80% of the data, then dropping slightly. In the mixed test set, the model showed a strong initial accuracy of 87% with 50% of the KTFE training data included, and the accuracy proceeded to vary across the different dataset sizes while remaining above 80%. The model peaked at 88% when the full dataset was used during the training process.

4.3.6 Discussion

The aim of this study was to evaluate the performance of different DL models in classifying emotions using thermal cameras. The performance of ResNet, Google ViT, and a modified ViT was compared, providing some interesting insights. The ResNet model, comprising a CNN architecture, displayed robust general performance, excelling particularly in classifying Fear and Surprise emotions with perfect precision and performing well in test instances involving the KTFE dataset. Meanwhile, both ViT models yielded slightly higher overall accuracies, with the modified ViT achieving the highest precision across all tests. Notably, the Google ViT achieved perfect scores for Anger and Happy emotions and demonstrated exceptional generalisation capabilities when tests included the KTFE dataset.

Although the ResNet architecture has been proven to be effective for FER [152] [153], the results in this work suggest that ViT models, especially the modified ViT, are particularly adept at thermal FER, potentially due to their architecture that enables more nuanced feature extraction [154]. The rapid learning process, portrayed in the results and overall performance of the ViT models, aligns with previous findings indicating their advantage in image classification tasks [155] [156] [157].

Using low-cost cameras and making thermal emotion classification more accessible could have a substantial impact in fields such as affective computing and healthcare. This data could then be used to train models and form part of advanced systems that monitor emotional well-being or enhance human-computer interaction by accurately interpreting emotional states in real-time settings. The results show that the generalisation capabilities provide the potential to capture wider variability and incorporate diverse data, which is pivotal for real-world applications where data conditions can widely vary. Moreover, as the thermal spectrum is considered non-invasive data [128], this work contributes meaningfully to developing ethical applications, where privacy and security risks are prevalent.

Despite the promising outcomes, this study does contain some constraints and limitations. One significant limitation is the dataset size and diversity. While the models performed well, the dataset does not contain any "in the wild" data [158], which might feature variations in head angles, head rotation, and variations in lighting or shadows. Also, while gender and ethnic diversity were not the main focus of this work, the dataset could be improved and considered more balanced with a better male-to-female ratio and increased ethnic diversity. However, this work shows that using this dataset to form a hybrid dataset can yield respectable results and offers a potential solution when only low-cost cameras are available, yet data collection remains difficult.

Future work should address these limitations by expanding the dataset, incorporating more diverse test participants, and including the collection of data where real-world conditions are replicated. Exploring hybrid datasets should also be a focus to combine state-of-the-art datasets with this low-cost hardware dataset to investigate the best ratio to include for optimal performance.

Additionally, exploring hybrid models that combine the strengths of convolutions and transformers may offer enhanced performance in specific circumstances, particularly where both local feature extraction and global context are valuable. Techniques such as transfer learning could be further utilised to enable models to maintain high performance across diverse environments without extensive retraining. In this chapter, we applied transfer learning by fine-tuning a pre-trained Google Vision Transformer model on our thermal dataset, demonstrating strong generalisation even with limited training data. Furthermore, convolutional methods such as region-based cropping or attention-guided feature enhancement could be employed to refine input representations in hybrid architectures.

Finally, while this work primarily contributes a novel dataset and comparative model evaluation, the findings have potential implications for real-world system integration. The rapid convergence and inference speed of the ViT models - particularly the modified version trained on smaller datasets - suggest viability for deployment on edge devices or embedded systems. Real-time metrics such as latency, throughput, and energy efficiency could be benchmarked in future implementations, especially in applications like driver monitoring, emotion-aware user interfaces, or mental health support tools [159–161]. Testing these models on platforms such as Raspberry Pi or NVIDIA Jetson would help identify hardware limitations and inform system architecture decisions under real-world constraints.

Although this dataset is relatively small in scale, it offers several advantages that support reliable model evaluation including moderate participant diversity across age and national background, which - along with its structured class distribution and controlled elicitation protocol - enhances its utility as a benchmark for evaluating models in low-cost, real-world emotion recognition scenarios.

4.4 Conclusions

This work introduces our thermal FER dataset and explores the potential of ViT models as a viable alternative to traditional CNNs for thermal emotion classification, particularly within the context of low-resolution imagery obtained from IoT-based low-cost thermal cameras. Two ViT models were used: one developed by Google and a modified model. Additionally, ResNet was used for comparison within the CNN architecture. Our findings highlight several key insights that contribute to the fields of emotion recognition technology and affective computing.

Compared to ResNet, the ViT models demonstrated a rapid learning capability, with the modified ViT model outperforming both ResNet and the Google model on our newly introduced dataset. Furthermore, our results indicate that diverse training data can significantly enhance model performance, as the models achieved high accuracy using a hybrid dataset, combining the KTFE dataset with our own. The Google ViT model yielded an overall accuracy of 94

In conclusion, these findings demonstrate the impact affordable hardware and state-of-the-art model architectures can have on emotion recognition. This work shows that leveraging low-cost, low-resolution cameras with advanced models like ViTs can make emotion recognition systems more accessible and scalable across varying applications. The study not only highlights the superiority of ViT models over traditional CNNs but also emphasises the role of diverse datasets in training ML models. In addition to its affordability, the dataset's structured composition, manual annotation process, and inclusion of participants from five national backgrounds provide a solid foundation for assessing FER models under controlled and reproducible conditions.

Further exploration and refinement of ViT models and low-cost hardware solutions may potentially revolutionise their application in real-world scenarios. Specifically, future work should investigate lightweight ViT architectures optimised for edge devices, explore hybrid models that integrate CNNs and transformers for improved efficiency, and evaluate deployment on embedded platforms to assess latency, power consumption, and inference speed under practical constraints. While ViT models are not applied in the subsequent multimodal setting, Chapter 5 builds directly on the findings of this chapter by incorporating thermal imagery - captured using the same low-cost hardware - into a real-time emotion recognition framework alongside speech and text. This transition shifts focus from model architecture comparison to modality integration and system performance, demonstrating the practical viability of thermal data in multimodal fusion strategies for edge deployment.

Chapter 5

Real-Time Multimodal Emotion Recognition

5.1 Introduction

Emotion recognition plays a vital role in enabling intelligent systems to interpret human affective states and respond accordingly, spanning a wide range of domains including human-computer interaction, mental health monitoring and autonomous vehicles, and assistive technologies [162–164]. Traditional approaches to emotion recognition have predominantly relied on unimodal data sources, such as speech signals, facial expressions, or textual content. While effective under constrained conditions, unimodal systems often struggle where emotional cues may be ambiguous, missing, or noisy [165].

MER, which leverages information from multiple data sources, has emerged as a promising direction to address these challenges [166]. By combining signals such as speech, facial expressions, and semantics, multimodal systems aim to achieve greater robustness and generalisability, however, effective fusion of modalities remains a significant research challenge. Differences in temporal alignment, data representation, and modality reliability can complicate the fusion process, and naive integration strategies may introduce noise or redundancy rather than improving performance [167].

Recent research has highlighted the potential of thermal imaging for emotion recognition [168]. Unlike conventional imagery from RGB cameras, thermal cameras capture physiological responses that correlate with emotional states [169]. Thermal imaging also offers advantages in low-light or visually challenging environments, making it a robust modality for affective computing applications [21]. Despite its promise, thermal imagery has been underexplored in the context of multimodal fusion alongside speech and text and prior work rarely integrated these modalities. Moreover, fusion strategies are often applied without thorough comparative analysis under deployment constraints, particularly in edge computing environments.

This chapter addresses these gaps and investigates the performance of an emotion recognition system that combines thermal images, speech, and text through both feature-level and decision-level fusion strategies. To model each modality effectively, we utilise CNNs and Long Short-Term Memory (LSTM) networks, leveraging their strengths in spatial and contextual feature learning. Using two well-known emotion recognition corpora - Interactive Emotional Dyadic Motion Capture (IEMOCAP) [35] and Multimodal EmotionLines Dataset (MELD) [170] - paired with thermal imaging datasets from UWSTF and KTFE [171], we systematically evaluate unimodal, bimodal, and tri-modal configurations. Furthermore, we assess the latency and throughput of the models to determine their viability for deployment on edge devices where computational resources are constrained. Our contributions are threefold: (1) we conduct a comprehensive evaluation of thermal, acoustic, and linguistic signals for emotion recognition across structured and conversational datasets; (2) we compare feature-level and decision-level fusion strategies in terms of accuracy, robustness, and generalisability; and (3) we present detailed system performance metrics, including inference latency and throughput, to inform practical deployment considerations.

The rest of this chapter is outlined as follows: Section 5.2 describes the proposed methodology, Section 5.3 details the results of each systematic evaluation, before concluding the work in Section 5.4.

5.2 Proposed Methodology

This section outlines the methodology adopted for building and evaluating a MER system. We begin by introducing the datasets used and the integration of these datasets into a unified framework. Following this, we detail the preprocessing techniques applied to each modality and the specific feature extraction methods used for thermal images, speech, and textual data. Finally, we present two multimodal data fusion strategies - feature-level fusion and decision-level fusion - used to combine the extracted features and improve emotion recognition performance.

5.2.1 Datasets

This study utilises a combination of thermal and multimodal datasets to enable the classification of emotions across different data modalities. Specifically, we include the UWSTF and KTFE thermal facial expression datasets for vision-based classification and the IEMOCAP and MELD datasets for speech and textual emotion classification. All four datasets were selected based on the presence of six shared core emotions: anger, fear, happiness, sadness, surprise, and neutrality.

UWSTF Dataset

The UWSTF dataset was introduced in our previous work exploring low-cost thermal FER. The dataset comprises images captured using the TOPDON TC001 thermal camera, which operates in the 8-14 Îijm spectral range with a resolution of 256ÃŮ192 pixels. Data was collected from 14 participants representing a range of ethnicities and age groups. The study protocol involved both spontaneous and posed emotion elicitation, using video clips and emoticon mimicking respectively. A total of 558 thermal images were manually extracted and labelled across six emotion classes and serves as a non-invasive and privacy-preserving resource for thermal FER model development. Examples of the dataset is shown in Figure 4.2, previewing each emotion class.

KTFE Dataset

The KTFE dataset [171] is a well-established thermal FER dataset featuring posed emotional expressions collected under controlled conditions. The dataset includes multiple emotion classes expressed by participants from Vietnamese, Japanese, and Thai backgrounds. Despite demographic limitations, KTFE provides high-resolution thermal imagery and reliable emotion annotations.

IEMOCAP Dataset

The IEMOCAP dataset [35] is a multimodal corpus frequently used in the field of speech emotion recognition. It contains approximately 12 hours of scripted and improvised dyadic interactions between professional actors. Each utterance is annotated with one of several emotion categories, including anger, fear, happiness, sadness, surprise, and neutral. The dataset includes aligned audio recordings and transcribed text, making it suitable for both speech and text-based emotion classification. IEMOCAP was selected to provide a controlled yet expressive speech and language dataset to support emotion modelling beyond the thermal modality.

MELD Dataset

The MELD dataset [170] is a benchmark corpus derived from the television series Friends, offering utterance-level annotations for seven emotions. For consistency with the rest of this study, only six overlapping emotions - anger, fear, happiness, sadness, surprise, and neutral - were retained. MELD features conversational exchanges and includes audio and text modalities, enabling the modelling of emotions in more natural, real-world settings. Compared to IEMOCAP, MELD introduces greater variability in speech patterns, background noise, and speaker dynamics, thereby enhancing model robustness in less constrained environments.

Dataset Splitting

All datasets were partitioned using a 70/10/20 split for training, validation, and testing, respectively. For the smaller UWSTF dataset, stratified sampling was applied to ensure balanced representation across the six emotion classes, with random selection used to align with class distributions. The KTFE dataset followed the same split strategy. For IEMOCAP, a speaker-independent approach was used by designating one session for testing and dividing the remaining sessions accordingly. MELD was used with its predefined splits, and only utterances containing one of the six target emotions and both audio and text modalities were retained. Random seeds were fixed across all splits to ensure reproducibility.

5.2.2 Preprocessing Steps

Prior to feature extraction and model training, the multimodal dataset underwent a series of preprocessing steps tailored to each data type, with the objective of preparing the raw data while preserving important emotional cues and reducing noise and variability. To accommodate the differing characteristics of each modality, dedicated pipelines were set up to extract features in a consistent and scalable format. The following subsections outline the modality-specific preprocessing techniques used in this study.

Image Features

For the image modality, all images underwent preprocessing including greyscale conversion, face detection using Haar cascades, background removal, resizing to 48x48, and normalisation. These inputs were fed directly into a CNN without applying any handcrafted feature extraction or dimensionality reduction. This design choice was made to allow the CNN to autonomously learn spatial hierarchies of features through its convolutional layers and to keep preprocessing steps to a minimum for an optimal real-time approach [39, 172].

Speech Features

Speech data was pre-processed using acoustic feature extraction methods that capture both static and dynamic properties of the audio signal, specifically, Mel-Frequency Cepstral Coefficients (MFCCs) and delta coefficients. This enabled features to represent the short-term power spectrum of the speech by capturing prosodic variations and temporal fluctuations indicative of underlying emotional states [173].

Text Features

The textual data was pre-processed using standard NLP techniques which included changing text to lowercase, removing stop words, word stemming, and removing punctuation before being transformed into numerical feature vectors using the TF-IDF method. This technique quantifies the importance of words relative to the corpus, enabling an informative sparse representation of speech utterances. A vectorizer was fitted to the dataset to capture the statistical distribution of terms across the corpus and subsequently used to transform all text samples. This approach is appropriate as it ensures that linguistically significant terms exert greater influence during model training while common semantically neutral words are down weighted. Also, as demonstrated in our previous work [174], TF-IDF was the most efficient feature set for text classification on edge.

5.2.3 Data Fusion

This study adopts and compares feature-level and decision-level fusion strategies, enabling the evaluation of methods that preserve the specialised learning capacity of each unimodal model while capturing cross-modal interactions through joint training. It also facilitates efficient deployment on edge devices while maximising the synergistic value of multimodal inputs.

Feature Level Fusion

In the feature-level component of the fusion pipeline each modality was processed independently through its respective model up to the penultimate layer. The output logits from the image, speech, and text models were then concatenated to form a composite feature vector. By combining the intermediate learned representations from each modality, the model gains access to richer, higher-level semantic and affective cues. Although the modality-specific models were trained independently, the final classification head was trained on the fused representation, allowing for cross-modal dependencies to be learned without retraining the base models. This strategy offers a favourable trade-off between computational efficiency and representational richness.

Decision Level Fusion

For decision-level fusion each unimodal model was trained independently and capable of producing its own predictions. This independence allows for flexible deployment scenarios where one or more modalities may be absent. While the final fusion model goes beyond traditional late fusion by enabling joint learning, the underlying architecture retains the modularity and resilience characteristic of decision-level strategies. Retaining independently trained modality-specific models preserves the robustness of late fusion, particularly in edge settings where sensor data may be missing or noisy, providing flexibility for selective modality activation, making it adaptable in real-world applications.

5.2.4 Models

The architecture design across modalities was informed by the need to maintain computational efficiency for deployment on edge devices with limited processing power as depicted in the comprehensive overview of model architectures in Figure 5.1. As such, each model was compiled to balance representational capacity with minimal parameter count. All models were compiled using the Adam optimiser and trained with the sparse categorical cross-entropy loss function.

To prevent overfitting and enable efficient learning the training process included early stopping and learning rate reduction based on the model performance on the evaluation data. Model checkpoints were also used to retain the best-performing weights throughout training. This section outlines the architecture used for each modality.

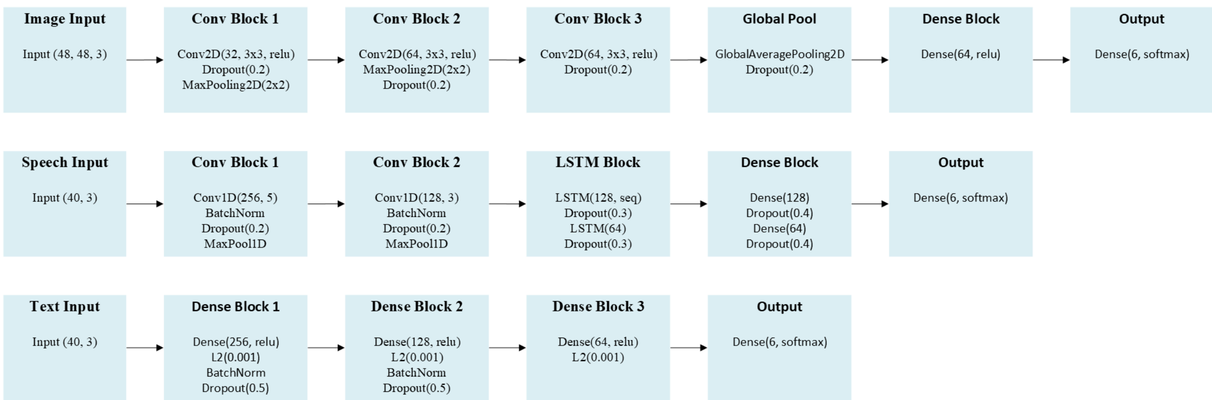


Figure 5.1: Overview of the unimodal model architecture

Image Model

For the image-based modality shown in Figure 5.1 a CNN was used to extract visual features from facial expressions. The architecture consisted of three convolutional layers with increasing filter sizes interleaved with max pooling operations to progressively reduce spatial dimensionality. A global average pooling layer was utilised to reduce the risk of overfitting by averaging spatial representations. This was followed by two fully connected layers: one with 64 units for feature integration and a final dense output layer with six units representing the target emotion classes. The architecture was lightweight and optimised for edge deployment with under 65,000 trainable parameters in total.

Facial expressions are known to carry rich affective information, but the computational cost of processing high-resolution images with deep CNNs can introduce latency on edge devices. The use of a compact CNN with global average pooling instead of flattening substantially reduced parameter count while preserving critical spatial information, making the model well-suited for efficient emotion recognition in real-time edge scenarios.

Speech Model

The model for the speech modality presented in Figure 5.1 integrated both convolutional and recurrent components to capture acoustic and temporal dynamic. The architecture began with two convolution blocks, followed by batch normalisation and dropout layers to enhance model stability. Max pooling layers were used to reduce the feature maps and retain dominant patterns.

The convolutional block was followed by a stacked bidirectional LSTM architecture to model temporal dependencies. The first LSTM layer returned sequences to feed into a second LSTM, which was then passed through a series of fully connected layers regularised with dropout and concluding with a SoftMax output layer. Although leading to slightly more trainable parameters than the image model at approximately 300,000, the architecture provides a balance between expressive power and computation efficiency.

A hybrid CNN-LSTM architecture was chosen because the speech data contains emotional content in both spectral characteristics and prosodic flow, enabling the model to learn time-aware acoustic patterns without requiring excessive depth. This allows the model to remain compact and fast enough for edge deployment.

Text Model

A shallow feed-forward neural network was designed to predict emotions from the high-dimensional TF-IDF sparse input as highlighted in Figure 5.1. The TF-IDF vectors were then passed through a series of dense layers for dimensionality reduction, with batch normalisation and dropout applied after the first two hidden layers to stabilise learning and reduce overfitting. Despite the high dimensionality of the input vector the network maintained a low computational footprint and generalised well due to its constrained depth and regularisation mechanisms.

TF-IDF is well-suited for environments with limited computational resources, hence the architecture was purposely shallow and chosen over embedding-based or transformer approaches to ensure scalability to edge boards while still learning semantic patterns from the text.

Multimodal Model

To integrate the complementary strengths of each modality, we implemented both early and late fusion strategies within a unified multimodal architecture, as illustrated in Figure 5.2. This architecture directly incorporates the unimodal models previously presented in Figure 5.1, with their outputs combined to form the multimodal system. In the late fusion setup, these sub-networks were first trained independently as unimodal classifiers, and their penultimate layer activations were then concatenated to form a fusion vector. This vector was passed through a fully connected dense layer, and a final output dense layer with SoftMax activation for classification. Figure 5.2 clearly shows this sequence, highlighting how the outputs of the unimodal networks are integrated and processed for final emotion prediction.

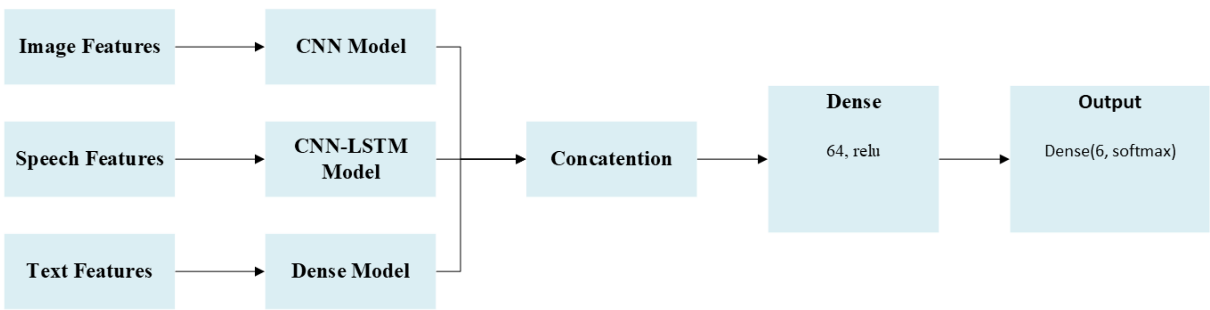


Figure 5.2: Overview of the multimodal model architecture

In contrast, the feature-level fusion approach involved combining the modality-specific features earlier in the network, allowing joint learning from the outset. The late fusion strategy was particularly beneficial for its modularity and flexibility, especially in edge computing scenarios where modality availability may vary at inference time. By pre-training unimodal components, the system preserved the strengths of each modality while combining through cross-modal integration at the decision level.

5.2.5 Real-Time System

To support real-time recognition in resource-constrained edge environments, the system was adapted to process sequential inputs for thermal images, speech, and text, simulating a live streaming environment. Instead of batch inference, each input was processed individually as it arrived. Speech and text data were handled using a sliding window approach, where a fixed-length segment was extracted for each inference step. For each speech or text window, a corresponding thermal image was selected using temporal alignment which was the facial frame closest in timestamp to the midpoint of the speech or text window.

The selected modalities were assembled into multimodal feature vectors and fed into either the feature-level or decision-level fusion models. Inference was performed on each new sample as it became available, without additional buffering beyond the current window, minimising memory usage and processing delays. This edge-oriented streaming setup enabled accurate measurement of latency and throughput under practical deployment constraints, demonstrating the system's capability to operate efficiently on low-power, resource-limited devices.

5.2.6 Metrics

To evaluate the performance of each model across unimodal and multimodal configurations accuracy and F1 score were employed to provide a balanced assessment of both overall correctness and class-wise predictive reliability. Accuracy measures the proportion of correctly classified instances over the total number of predictions and provides a general indication of model performance. While accuracy is useful for gauging global predictive performance, it may not fully reflect a model's ability to identify minority classes in imbalanced datasets.

To address this the F1 score was also calculated which is the harmonic mean of precision and recall, offering a more nuanced view of classification effectiveness, especially for underrepresented emotion classes. It balances the trade-off between false positives and false negatives, making it a more informative metric in scenarios where uneven class distribution may affect standard accuracy. Both metrics were computed for each model and feature representation, providing a comprehensive basis for comparison across modalities and fusion strategies.

5.3 Results

This section presents the experimental results for unimodal and MER across thermal image, speech, and text modalities. All models were evaluated using standard classification accuracy and macro-averaged F1-score metrics.

5.3.1 Unimodal Emotion Recognition

The results of unimodal emotion recognition experiments using thermal image, speech, and text modalities are reported across the four datasets. Figure 5.3 reveals the accuracy and macro F1-scores for each modality - dataset pair, with a further detailed analysis in subsequent sections.

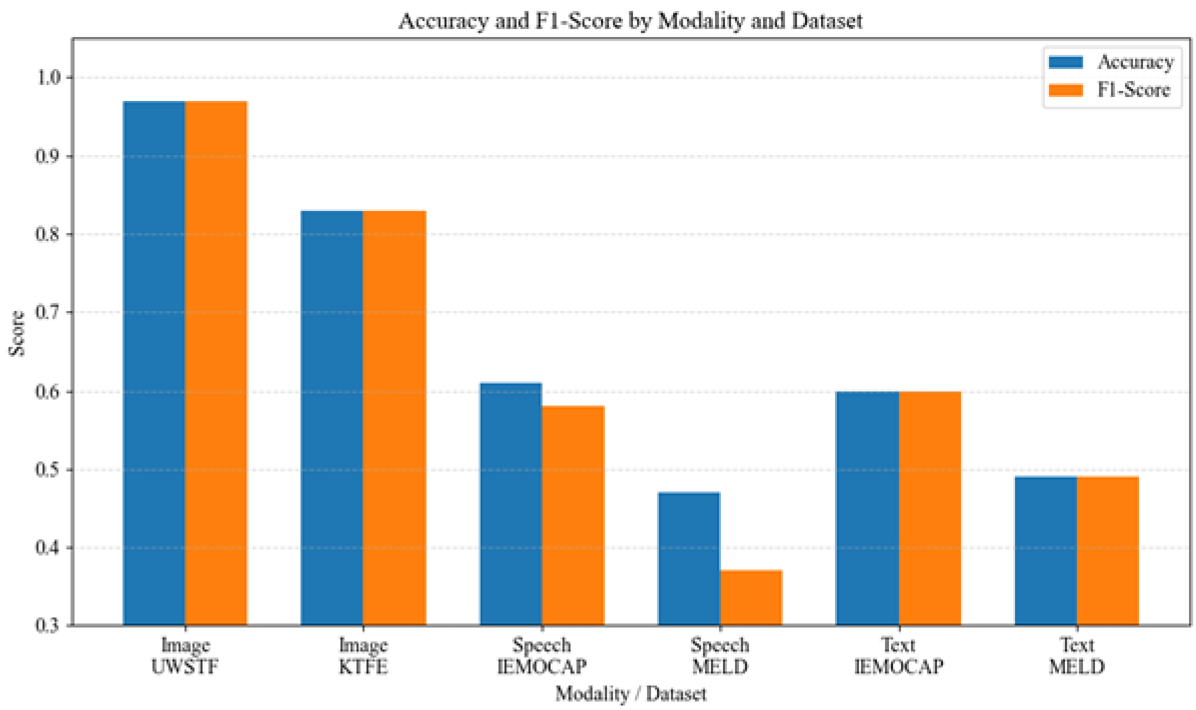
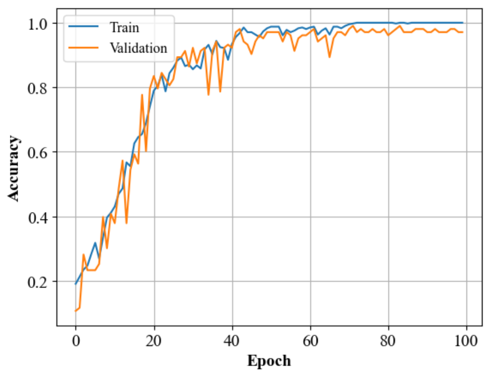


Figure 5.3: Summary of unimodal classification performance

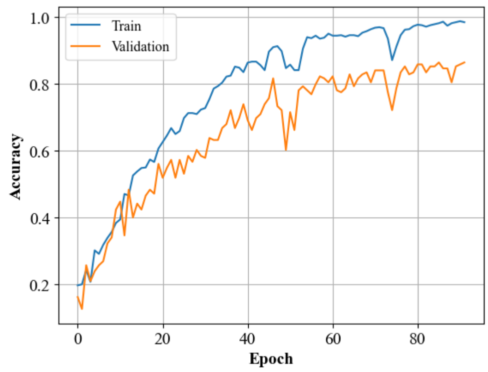
Image Modality

Thermal image-based emotion classification yielded the highest unimodal performance, particularly on the UWSTF dataset. The model achieved an accuracy of 97% and a macro F1-score of 0.97. Per-class performance was consistently high with several classes reaching perfect precision and recall. The training history, shown in Figure 5.4a, indicated a steady increase in accuracy, converging rapidly after 60 epochs and achieving near-perfect performance by epoch 80. Validation accuracy stabilised at 98-100%, indicating excellent generalisation and minimal overfitting.

Performance on the KTFE dataset was respectively lower, however, with 83% accuracy and a macro F1-score of 0.83. Class-level results varied more significantly, with the lowest F1-score observed for anger, reflecting difficulty in classifying certain expressions. The training trajectory, displayed in Figure 5.4b, showed a similar convergence pattern, albeit with slightly more fluctuation in validation accuracy. Final validation performance plateaued around 86%, with peak accuracy reaching 87%.



(a) Training history for UWSTF dataset



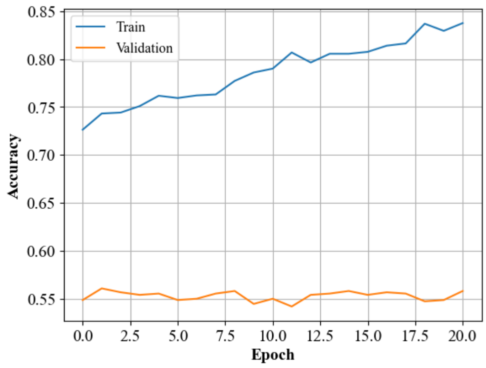
(b) KTFE training history

Figure 5.4: Training history for the UWSTF and KTFE datasets

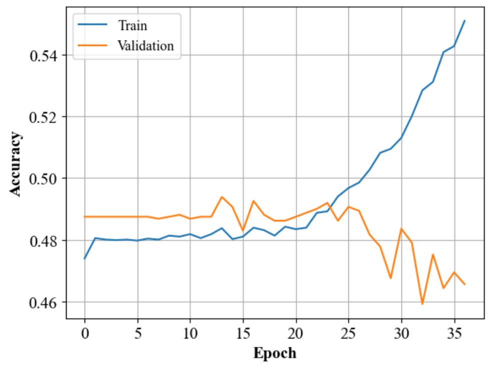
Speech Modality

Speech-based recognition showed moderate to low performance across datasets. On IEMOCAP, the model achieved 61% accuracy with an F1-score of 0.58. High variance across class-level metrics was observed with some classes performing better than others, the neutral and anger classes performing the best in this instance achieving F1-scores above 0.60.

The results indicate the models' experienced challenges in distinguishing certain emotions based solely on prosodic or acoustic features. The training curve in Figure 5.5a indicated stable convergence, but validation performance plateaued early at 56%.



(a) Training history for IEMOCAP speech dataset



(b) MELD speech training history

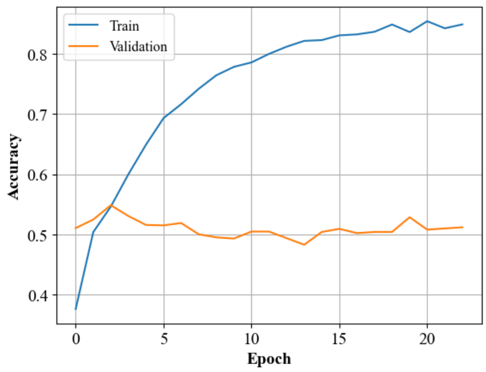
Figure 5.5: Training history for the IEMOCAP and MELD speech datasets

On MELD, performance further declined to 47% accuracy and an F1-score of 0.37. Although minority classes were poorly classified, in this instance - similar to the IEMOCAP dataset, the neutral class performed best with an F1-score of 0.65. Training accuracy showed a gradual increase, as shown in Figure 5.5b, but validation accuracy remained largely stagnant, never exceeding 49%. This is consistent with the more complex, spontaneous nature of MELD conversations and the acoustic variability present in multi-party dialogues.

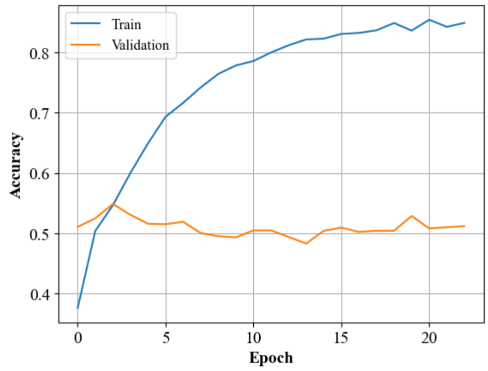
Text Modality

Text-based emotion recognition exhibited results comparable to speech on IEMOCAP, with 60% accuracy and a macro F1-score of 0.60. Also similar to the speech model, dominant classes such as neutral and anger performed the best, achieving F1-scores of 0.61 and 0.67 respectively, with performance on smaller classes weaker. Training history, presented in Figure 5.6a, revealed strong convergence after 10 epochs, with validation accuracy stabilising around 64%.

Performance declined on the MELD dataset, with 49% accuracy and a macro F1-score of 0.46. This dataset's linguistic complexity and informal structure introduced classification difficulty. The classifier struggled with under-represented classes such as fear and



(a) Training history for IEMOCAP text dataset



(b) MELD text training history

Figure 5.6: Training history for the IEMOCAP and MELD text datasets

sadness, and although training accuracy approached 85%, validation accuracy peaked at 53%. The disparity between training and validation performance was more evident than in IEMOCAP, as shown in Figure 5.6b, highlighting the model had trouble when generalising to noisy, conversational text.

5.3.2 Multimodal Emotion Recognition

This section presents the results of the multimodal part of the analysis using both feature-level fusion and decision-level fusion strategies and investigates performance trends and learning behaviour for both fusion strategies. The results include experiments across the four datasets described under three fusion configurations: image and speech, image and text, and the full tri-modal combination. Performance is assessed using classification accuracy and macro-averaged F1-score. Table 5.1 summarises the results across all configurations.

Fusion Type	**Dataset Pair**	**Accuracy (Early / Late)**	**Macro F1-Score (Early / Late)**
Image + Speech	UWSTF + IEMOCAP	56 / 99	0.50 / 0.99
	KTFE + IEMOCAP	60 / 87	0.51 / 0.86
	UWSTF + MELD	59 / 98	0.53 / 0.99
	KTFE + MELD	40 / 89	0.38 / 0.88
Image + Text	UWSTF + IEMOCAP	57 / 96	0.56 / 0.94
	KTFE + IEMOCAP	63 / 87	0.64 / 0.85
	UWSTF + MELD	62 / 98	0.55 / 0.98
	KTFE + MELD	46 / 89	0.46 / 0.88
Image + Speech + Text	UWSTF + IEMOCAP	61 / 97	0.61 / 0.97
	KTFE + IEMOCAP	65 / 88	0.66 / 0.88
	UWSTF + MELD	53 / 97	0.45 / 0.97
	KTFE + MELD	40 / 88	0.39 / 0.88

Table 5.1: Fusion performance for each dataset pair and fusion type

Image and Speech Fusion

In the feature-level fusion setup, where image and speech features were concatenated and learned jointly, performance was moderate across datasets. The best result was observed with the KTFE + IEMOCAP pair, which achieved 60% accuracy and F1-score of 0.51. The model demonstrated strong recall in specific classes, notably the fear class, though this came at the cost of overall precision and class balance. The UWSTF + IEMOCAP configuration yielded slightly lower performance with high class-wise variability.

Experiments involving MELD were less successful. Feature-level fusion in UWSTF + MELD resulted in 59% accuracy and KTFE + MELD achieved only 40% accuracy. Despite achieving strong training accuracy, the validation performance was inconsistent, particularly for underrepresented classes. By contrast, the decision-level fusion models - where image and speech were processed independently and merged at the output stage - consistently outperformed their feature-level counterparts. The UWSTF + IEMOCAP model achieved 99% accuracy and a macro F1-score of 0.99, with near-perfect class-level metrics and rapid convergence. KTFE + IEMOCAP also performed well, with 87% accuracy.

Similarly, UWSTF + MELD and KTFE + MELD reached 98% and 89% accuracy respectively, highlighting the robustness of thermal imagery in compensating for variability in speech data. Training histories for decision-level fusion models indicated strong generalisation with minimal overfitting, even in more challenging dataset pairings.

Overall, decision-level fusion yielded superior performance in both accuracy and F1-score, particularly in configurations involving MELD. Feature-level fusion offered a modest gain over unimodal baselines but struggled to match the robustness and class consistency of the decision-level approach.

Image and Text Fusion

In the feature-level fusion experiments, results were stronger than those observed for image and speech combinations. The best performance was seen in the KTFE + IEMOCAP configuration, achieving 63% accuracy and a macro F1-score of 0.64. The model effectively distinguished between emotional categories, with F1-scores exceeding 0.75 in anger and fear classes. Training converged reliably, and validation accuracy remained above 65% in the later stages.

The UWSTF + IEMOCAP model achieved 57% accuracy and an F1-score of 0.56. Performance was imbalanced across classes - the surprise class was well recognised while the sad class lagged, likely due to textual ambiguity or imbalance. Nevertheless, training and validation stability were maintained throughout.

On MELD, performance declined. The UWSTF + MELD configuration reached 62% accuracy while KTFE + MELD dropped to 46% accuracy. Class-wise precision was highest in the surprise class but confusion remained in others, particularly in the anger class.

In contrast, decision-level fusion of image and text showed strong and more consistent performance. The UWSTF + IEMOCAP model achieved 96% accuracy and a macro F1-score of 0.94, with minor dips in recall on underrepresented classes. The model converged quickly, maintaining stable validation accuracy above 90%.

The KTFE + IEMOCAP setup performed similarly well, with 87% accuracy and an F1-score of 0.85, improving on the classification of speech fusion. On MELD, UWSTF +

MELD maintained high performance with 98% accuracy while KTFE + MELD achieved 89% accuracy. These results highlight the effectiveness of thermal imagery in disambiguating emotional context, even when the text modality alone is less reliable. Together, the findings confirm that text is a highly complementary modality when paired with thermal images, and that decision-level fusion consistently outperforms feature-level fusion across all configurations.

Image, Speech, and Text Fusion

In the feature-level fusion setting for image, speech and text fusion, performance was generally mixed. The highest accuracy was observed in the KTFE + IEMOCAP configuration, where the model achieved 65% accuracy and a macro F1-score of 0.66 - slightly higher than its bimodal counterparts. The UWSTF + IEMOCAP model followed closely, with 61% accuracy and an F1-score of 0.61, demonstrating balanced class-wise performance but only marginal gains over bimodal fusion.

Performance dropped considerably with MELD. The UWSTF + MELD setup achieved 53% accuracy while KTFE + MELD reached only 40% accuracy. While training accuracy remained high, validation scores were erratic, highlighting the difficulty the models had when conducting joint feature learning across three heterogeneous modalities.

By comparison, decision-level fusion of all three modalities yielded high but not consistently superior performance over simpler configurations. The UWSTF + IEMOCAP model reached 97% accuracy with a macro F1-score of 0.97, maintaining class consistency but offering no substantial improvement over bimodal results. Similarly, KTFE + IEMOCAP reached 88% accuracy, echoing earlier trends where added modalities contributed minimally beyond a certain threshold.

Tri-modal fusion on MELD also followed this pattern. Both UWSTF + MELD and KTFE + MELD configurations achieved 97% and 88% accuracy respectively, mirroring the bimodal performance. Although training and validation curves indicated strong generalisation, the additional modality did not translate into performance gains.

In summary, while tri-modal fusion offered robustness and class stability, it did not consistently outperform bimodal models. These results support prior findings that adding modalities can enhance model resilience but may lead to diminishing returns without careful integration, weighting, and synchronisation across modalities.

5.3.3 Edge Performance

To assess the feasibility of deploying the MER system on edge devices, inference latency and throughput was measured for both feature-level and decision-level fusion models using a Raspberry Pi model 4B. Additionally, preprocessing times for individual modalities were recorded to understand the full end-to-end system performance.

Latency and throughput were evaluated using a custom performance measurement tool written in Python which allowed recording single-sample inference time and throughput across batches. Inference latency measures the time taken for a single forward pass through the model, while throughput captures the number of samples processed per second. Preprocessing time was separately measured for each modality to quantify the data preparation overhead. For all throughput experiments, a batch size of 32 was used over a total of 104 samples, resulting in three batches per test.

The results, summarised in Table 5.2, show that decision-level fusion models exhibited longer inference times compared to feature-level fusion models. Specifically, inference for the hybrid image-speech-text configuration required 1.61 seconds, while the feature-level fusion version of the same configuration completed in 0.78 seconds. Similar trends were observed across the image-speech and image-text pairings, where early fusion consistently demonstrated lower latency.

Model	**Inference Latency (s)**	**Inference Throughput (samples/s)**
Decision-Level Fusion Image-Speech-Text	1.61	4272.17
Decision-Level Fusion Image-Speech	0.68	4424.47
Decision-Level Fusion Image-Text	0.62	4669.61
Feature-Level Fusion Image-Speech-Text	0.78	4308.90
Feature-Level Fusion Image-Speech	0.61	4779.81
Feature-Level Fusion Image-Text	0.54	4702.63

Table 5.2: Edge inference latency and throughput

Throughput results, however, remained high across all configurations. Decision-level models achieved throughput rates of 4272.17, 4424.47, and 4669.61 samples per second for the image-speech-text, image-speech, and image-text configurations, respectively. Feature-level fusion models exhibited slightly higher throughput, with the feature-level image-speech configuration reaching 4779.81 samples per second.

Preprocessing times also varied across modalities as presented in Table 5.3. Preprocessing a single text input required only 0.01 seconds, making text the fastest modality to prepare. In contrast, thermal image preprocessing took 0.06 seconds per sample, and speech preprocessing was the most time-consuming at 0.08 seconds per sample, largely due to the need for audio decoding and feature extraction. The corresponding preprocessing throughput rates were 673.12 samples per second for images, 858.93 samples per second for audio, and 514.10 samples per second for text.

Modality	Preprocessing Time per Sample (s)	Preprocessing Throughput (samples/s)
Image Preprocessing	0.06	673.12
Speech Preprocessing	0.08	858.93
Text Preprocessing	0.01	514.10

Table 5.3: Edge preprocessing latency and throughput

Despite the increased complexity introduced by tri-modal fusion, all configurations demonstrated acceptable inference speeds suitable for real-time or near-real-time edge deployment. These results confirm that with appropriate optimisation, particularly at the preprocessing stage, the proposed multimodal system can be effectively deployed in resource-constrained environments without substantial sacrifices in response time.

To further assess deployment feasibility under real-time streaming conditions, the inference pipeline was adapted to process data sequentially, simulating a live edge environment. Unlike batch-based processing, each input sample was handled individually as it arrived, reflecting real-world streaming scenarios for emotion recognition at the edge.

Speech and text inputs were processed using a sliding window approach, with corresponding thermal images temporally aligned to each window's midpoint. Inference was performed on each new sample without additional buffering, allowing accurate measurement of real-time latency and throughput under continuous streaming conditions.

The average real-time inference latency for the decision-level fusion model was 0.22 seconds per sample, yielding a throughput of 4.66 samples per second. The feature-level fusion model performed slightly better, with an average latency of 0.21 seconds per sample and a throughput of 4.70 samples per second. These results demonstrate that the system meets real-time constraints, achieving processing rates comfortably above 4 frames per second. The marginal latency difference between fusion strategies further suggests that both are suitable for real-time edge deployment when data is synchronised and streamed effectively.

5.3.4 Discussion

The results from both feature-level and decision-level fusion experiments provide several key insights into the role of modality combinations, dataset structure, fusion strategies, and system-level performance in MER.

First, dataset structure emerged as a crucial factor in model performance. Across both fusion strategies, configurations using IEMOCAP consistently outperformed those using MELD, highlighting the benefit of structured, balanced data with well-defined emotional categories. This trend was particularly evident in text and tri-modal fusion scenarios, where MELD's conversational informality and variability presented a consistent challenge.

Second, the choice of modality significantly affected results. Among bimodal configurations, image and text fusion generally outperformed image and speech combinations in both F1-score and training stability. This may reflect the richer semantic content and emotional cues embedded in language, particularly in structured settings, compared to the paralinguistic ambiguity of spontaneous speech.

Third, the addition of a third modality did not consistently translate into improved performance. In many cases, tri-modal models performed similarly to or only slightly better than their bimodal counterparts. This was observed across both early and late fusion setups and suggests a saturation point in model capacity, beyond which added complexity introduces redundancy or noise and may even reduce generalisability without appropriate alignment or weighting mechanisms. Noisy or less informative modalities, such as conversational text and speech from MELD, may have introduced conflicting signals, further limiting the benefits of tri-modal fusion.

The fusion strategy itself also played a critical role. Feature-level fusion showed greater variability across configurations and tended to underperform relative to decision-level fusion. This can be attributed to the inherent difficulty of jointly learning from heterogeneous feature spaces without prior modality-specific optimisation. In contrast, decision-level fusion allowed models to specialise in processing their respective input domains before combining outputs, leading to higher accuracy and more robust generalisation across datasets.

From a unimodal perspective, thermal imaging emerged as the most reliable standalone modality. It achieved 97% accuracy on UWSTF and maintained high performance across all fusion combinations. On the other hand, unimodal speech and text models demonstrated limited performance, particularly on MELD, where emotion is often contextually implied rather than explicitly expressed.

Nevertheless, speech and text retained significant value in multimodal fusion, particularly when combined with thermal imagery. Fusion experiments demonstrated that pairing either modality with images consistently improved over unimodal baselines. For example, UWSTF + MELD achieved 98% accuracy using both image-speech and image-text decision-level fusion configurations, highlighting the complementary nature of the modalities where thermal imaging captures facial expressions while speech and text modalities capture contextual and semantic information.

From a deployment perspective, latency and throughput analyses further clarified the trade-offs between fusion strategies. Feature-level fusion models consistently achieved lower inference latency and higher throughput compared to decision-level fusion. For instance, feature-level fusion of image-speech-text required 0.78 seconds per inference, compared to 1.61 seconds for the decision-level model.

Preprocessing times varied significantly across modalities, with text requiring minimal preparation while audio preprocessing was the slowest due to additional feature extraction overhead. These findings suggest that although tri-modal configurations offer robustness, the minor performance gains compared to bimodal setups may not justify the increased computational complexity, particularly in resource-constrained environments.

Despite these encouraging results, this study has several limitations that should be acknowledged. First, the video keyframe selection process for thermal image extraction was not fully optimised, which may have impacted the quality and representativeness of the visual data. Similarly, data preprocessing pipelines were designed for general applicability rather than modality-specific optimisation, potentially limiting model performance, particularly for speech-based inputs. The speech models also exhibited lower performance compared to text and image modalities, suggesting a need for more advanced acoustic feature engineering or alternative model architectures. Additionally, due to the limited availability of large, aligned multimodal datasets, this study combined different image and speech datasets, which may introduce dataset bias and affect generalisation. However, this approach is consistent with practices in related studies addressing similar data scarcity challenges.

Overall, the findings support the feasibility of deploying MER systems on edge devices, particularly using decision-level fusion of thermal imagery with either speech or text. However, system design should prioritise simplicity, efficiency, and alignment between modalities over maximising input complexity, in order to balance accuracy, responsiveness, and practical usability.

5.4 Conclusion

This chapter investigated the performance of MER systems combining thermal images, speech, and text modalities, with a specific focus on their suitability for deployment in resource-constrained edge environments. Building on the findings from Chapter 3, which identified efficient textual feature representations suitable for edge deployment, and Chapter 4, which demonstrated the effectiveness of thermal facial expression recognition using low-cost hardware, this chapter integrated these modalities into a real-time fusion framework. While the ViT models explored in Chapter 4 were not included here due to computational demands, the thermal data collected using the same low-cost IoT device formed a core component of the multimodal system. Both feature-level and decision-level fusion strategies were evaluated across structured and conversational datasets, revealing key insights into the impact of modality selection, fusion design, and dataset quality on overall system performance and deployability.

Thermal imagery consistently provided the strongest unimodal signals, outperforming speech and text modalities. When modalities were fused, both image-speech and image-text combinations substantially improved accuracy over unimodal baselines. Decision-level fusion models achieved the highest performance, with accuracy reaching up to 99% on structured datasets. Feature-level fusion models, while more computationally efficient, demonstrated greater sensitivity to dataset variability and modality imbalance.

Tri-modal configurations incorporating all three modalities did not consistently outperform bimodal models, indicating that simply increasing modality count leads to diminishing returns unless supported by carefully optimised fusion architectures. Importantly, system-level evaluations confirmed that all multimodal configurations maintained low inference latency and high throughput on edge devices such as the Raspberry Pi 4B, validating their feasibility for real-time, offline emotion recognition. The trade-off between computational efficiency and classification accuracy was particularly evident in feature-level fusion models, which are well-suited for low-power, latency-sensitive edge deployments.

Overall, the results highlight that effective MER hinges not only on the number of modalities used, but also on their integration quality, dataset structure, and, critically, the strategic balance between system complexity and the limitations of edge hardware in terms of processing power, memory footprint, and energy efficiency. Future work will explore adaptive fusion strategies that dynamically weight modalities based on context, and further optimise preprocessing pipelines to support scalable, real-time emotion recognition on embedded and edge AI platforms.

Chapter 6

Conclusion and Future Work

Conclusion

This research has successfully achieved its stated objectives and contributed novel, practical advancements in the field of emotion recognition. Specifically, it delivered:

- a comprehensive evaluation of unimodal models using thermal, speech, and text modalities.
- a reproducible framework for low-cost, multimodal data collection.
- the design and deployment of a real-time, resource-efficient MER system capable of operating on edge devices.

Experimental results demonstrated that thermal imaging, collected using an IoT-based low-resolution camera, provided the most consistent and reliable unimodal performance, achieving classification accuracy up to 96% and outperforming traditional RGB-based methods in privacy-sensitive and resource-constrained conditions. Textual emotion classification using TF-IDF and SGD achieved up to 90% accuracy, offering a computationally efficient alternative to transformer-based models on embedded platforms. Speech and text modalities added valuable context but showed reduced robustness when used independently, particularly in noisy or ambiguous environments.

The developed multimodal framework integrated thermal, acoustic, and linguistic signals using both feature-level and decision-level fusion strategies. Decision-level fusion consistently delivered the highest performance, reaching up to 99% classification accuracy on structured datasets, and demonstrating strong generalisation across modality combinations. The system maintained real-time capability on a Raspberry Pi 4B, achieving inference latency as low as 0.22 seconds per sample and throughput exceeding 4,200 samples per second. These results confirm that accurate, scalable, and privacy-conscious emotion recognition is achievable on low-cost, resource-constrained platforms.

Overall, this work provides a validated, deployable MER system, bridging the gap between research and practical implementation. It lays the foundation for future advancements in affective computing, with direct applications in healthcare, education, assistive technology, and beyond.

Future Work

With the successful demonstration of a real-time, low-cost multimodal emotion recognition MER system on edge hardware, several key avenues for future research have emerged that could extend the contributions of this thesis and address identified limitations.

- **Preprocessing Optimisations:** Future research should focus on further optimising preprocessing pipelines, particularly for the speech and thermal modalities. Improvements in data preparation processes could significantly reduce system latency and enhance model performance, enabling more efficient real-time operation, especially under noisy and dynamic conditions.
- **Development of a New Synchronous Dataset:** One limitation identified in this work was the absence of a dataset where thermal, speech, and text data are captured in a fully synchronised manner. Future efforts should prioritise the creation of such a dataset to facilitate more effective modality fusion and alignment, supporting the development of models that better reflect real-world human interactions.
- **Key Video Frame Selection for Real-Time Analysis:** To further reduce computational overhead and improve processing efficiency, future research should explore

advanced techniques for key video frame selection. By intelligently identifying the most informative frames, or utilising a collection of frames, it will be possible to improve real-time analysis performance without compromising accuracy, which is critical for deployment on edge devices.

- **Exploration of Hybrid Fusion Models and Transfer Learning:** Investigating hybrid models that combine the strengths of CNNs and ViTs may further enhance model robustness and accuracy. Additionally, the application of transfer learning on low-resource platforms should be explored to improve model adaptability and reduce the need for extensive retraining in varied deployment environments.

By addressing these directions, future work can build on the validated foundation established in this thesis and further advance the development of scalable, robust, and context-aware emotion recognition systems for deployment in real-world environments.

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